



GATEWAY CLASSES

DSTL



Discrete Mathematics

ONE SHOT

Unit-IV

Algebraic Structures



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New channel

Gateway Classes official AKTU

Section – A

i. Define normal subgroup.

Section – C

3. Attempt any one part of the following

(a) Let $G = \{1, -1, i, -i\}$ with the operation of ordinary multiplication on G be an algebraic structure, where $i = \sqrt{-1}$.

(i) Determine whether G is Abelian .

(ii) Determine the order of each element in G .

(iii) Determine whether G is a cyclic group, if G is a cyclic group, then determine the generator/generators of the group G .

(iv) Determine a subgroup of the group G .

(b) Let $(G, *)$ and $(G', *)$ be any two groups and let e and e' be their respective identities. If f is a homomorphism of G into G' , then prove that (i) $f(e) = e'$ (ii) $f(X^{-1}) = [f(X)]^{-1}$, $\forall X \in G$

7. Attempt any one part of the following

(a) Prove that the set of residues $F = \{0, 1, 2, 3, 4\}$ modulo 5 is a field w.r.t. addition and multiplication of residue classes modulo 5. i.e. $(F, +_5, \times_5)$ is a field.

Section – A

- c. State and justify "Every cyclic group is an abelian group" .*
- d. State Ring and Field with example.*

Section – B

- b. Explain Cyclic group. Let H be a subgroup of a finite group G . Justify the statement "the order of H is a divisor of the order of G ".*

Section – C

- 4. Attempt any one part of the following*

(a) Define the binary operation $$ on $\mathbb{Z}$ by $x * y = x + y + 1$ for all x, y belongs to set of integers.*

*Verify that $(\mathbb{Z}, *)$ is an abelian group? Discuss the properties of abelian group.*

*(b)(i) Justify that "The intersection of any two subgroup of a group $(G, *)$ is again a subgroup of $(G, *)$ ".*

(ii) Justify that "If a, b are the arbitrary elements of a group G then $(ab)^2 = a^2b^2$

if and only if G is abelian.

Section – A

c. Define group.

d. Define ring.

Section – B

b. State and prove Lagrange theorem for group.

Section – C

4. Attempt any one part of the following

(a) A subgroup H of a group G is a normal subgroup if and only if $g^{-1}hg \in H$ for every $h \in H$ and $g \in G$.

(b) In a group $(G, *)$ prove that

(i) $(a^{-1})^{-1} = a$

(ii) $(ab)^{-1} = b^{-1}a^{-1}$

Section – A

- c. Let Z be the group of integers with binary operation $*$ defined by $a * b = a + b - 2$ for all $a, b \in Z$. Find the identity element of the group $\langle Z, * \rangle$
- d. Show that every cyclic group is abelian.

Section – B

- b. What do you mean by cosets of a subgroup? Consider the group Z of integers under addition and the subgroup $H = \{ \dots, -12, -6, 0, 6, 12, \dots \}$ consisting of multiple of 6
- (i) Find the cosets of H in Z
- (ii) What is the index of H in Z .

Section – C

4. Attempt any one part of the following
- (a) What is Ring? Define elementary properties of Ring with example.
- (b) Prove or disprove that intersection of two normal subgroups of a group G is again a normal subgroup of G .

Section – A

b. Define Ring and Field.

Section – B

- b. (i) Let H be a subgroup of a finite group G . Prove that order of H is a divisor of order of G .
(ii) Prove that every group of prime order is cyclic.

Section – C

4. Attempt any one part of the following

- (a) Let $G = \{1, -1, i, -i\}$ with the binary operation multiplication be an algebraic structure, where $i = \sqrt{-1}$. Determine whether G is an Abelian or not.
- (b) What is meant by a ring? Give examples of both commutative and non – commutative rings.

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

DISCRETE MATHEMATICS

UNIT -4 : Algebraic Structures

Lecture - 01

Today's Target

- *Binary Operations*
- Definition of Algebraic Structures
- PYQ
- DPP

UNIT -4 : Algebraic Structures

- **Definition**
- **Groups, Subgroups and order**
- **Cyclic groups**
- **Cosets**
- **Lagrange's theorem**
- **Normal subgroups, permutation and symmetric groups,**
- **Group Homomorphism**
- **Definition and elementary properties of rings and fields.**

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Binary Operation ($*$)

Binary Operation Combine two elements of the set to produce another element of the same set.

OR

Let G be a non empty set and $G \times G = \{(a, b) : a, b \in G\}$ then a function $*$: $G \times G \longrightarrow G$ defined by

$$* (a, b) = a * b \in G$$

is called a binary operation on the set G

Binary Operators

$$+, -, \times, \div$$

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Some Standard Sets

N : The set of natural numbers

W : The set of whole numbers

Z or I : The set of integers

Q : The set of rational numbers

R : The set of real numbers

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Example : 1 The operation $+$ on the set of natural number N is a binary operation.

$$+ : N \times N \longrightarrow N$$

$(N, +)$

$$+(a, b) = a + b \in N \quad \forall a, b \in N$$

$$+(1, 4) = 1 + 4 = 5 \in N$$

$$+(19, 51) = 19 + 51 = 70 \in N$$

Hence, $+$ is a binary operation on the set of natural number (N)

Example : 2 The operation $-$ on the set of natural number N is not a binary operation.

$$- : N \times N \longrightarrow N$$

$$(N, -) \times$$

$$-(a, b) = a - b \notin \forall a, b \in N$$

$$-(3, 1) = 3 - 1 = 2 \in N$$

$$-(1, 4) = 1 - 4 = -3 \notin N$$

$$-(5, 10) = 5 - 10 = -5 \notin N$$

Hence, $-$ is not a binary operation for the set of natural number (N)

Example : 3 The operation $-$ on the set of integers $\mathbb{Z}$ is a binary operation.

$$- : \mathbb{Z} \times \mathbb{Z} \longrightarrow \mathbb{Z}$$

$(\mathbb{Z}, -)$

$$-(a, b) = a - b \in \mathbb{Z} \quad \forall a, b \in \mathbb{Z}$$

$$-(2, 4) = 2 - 4 = -2 \in \mathbb{Z}$$

$$-(6, 3) = 6 - 3 = 3 \in \mathbb{Z}$$

$$-(-4, -2) = -4 + 2 = -2 \in \mathbb{Z}$$

Hence, $-$ is a binary operation for the set of integers

Example : 4 The operation $\times$ on the set of Real number R is a binary operation. $(*)$

$$\times : R \times R \rightarrow R$$

$$(R, \times)$$

$$\times(a, b) = a \times b \in R, \forall a, b \in R$$

$$\times(-2, 3) = -2 \times 3 = -6 \in R$$

$$\times(2.3, -4) = 2.3 \times (-4) = -9.2 \in R$$

Hence, $\times$ is a binary operation for the set of Real numbers

A non empty set G together with one or more binary operation is called an algebraic structure.

Examples

$(\mathbb{N}, +)$ $(\mathbb{Z}, -)$ $(\mathbb{R}, \times)$ $(\mathbb{R}, +, \times)$ $(\mathbb{R}, +, -, \times)$ etc

Properties of Binary Operations

1. Closure property
2. Associative property
3. Identity property
4. Inverse property
5. Commutative property

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Note

1. Closure = Algebraic Structure

2. Closure + Associative = Semi-groups

3. Closure + Associative + Identity = Monoids

4. Closure + Associative + Identity + Inverse = Groups

5. Closure + Associative + Identity + Inverse + Commutative = Abelian Groups

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Topic : Binary Operations

Q.1 The operation $+$ on the set of real number R is a binary operation.

Q.2 The operation $-$ on the set of real number R is a binary operation.

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DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

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UNIT -4 : Algebraic Structures

Lecture - 02

Today's Target

- *Properties of Binary Operation*
- *Different types of algebraic structures*
- PYQ
- DPP

(1) Closer Property

G is closed under the operation $*$ if

For all

$$\forall a, b \in G \quad \text{then } a * b \in G$$

(2) Associative Property

The operation $*$ on G is associative if

$$\forall a, b, c \in G$$

we have

$$(a * b) * c = a * (b * c)$$

(3) Identity Property (Existence of identity element)

The operation $*$ on G has an identity property if

$$\forall a \in G \quad \exists \text{ an element } e \in G \text{ such that}$$

$$a * e = a = e * a$$

Where $e = \text{identity element}$

(4) **Inverse Property** (Existence of inverse element)

The operation $*$ on G has an inverse property if

$\forall a \in G \quad \exists$ an element $b \in G$ such that

$$a * b = b * a = e$$

Where $b = \text{Inverse of } a$

(5) **Commutative Property**

The operation $*$ on G is commutative if

$\forall a, b \in G$

we have

$$a * b = b * a$$

Properties of Binary Operations

1. Closure property
2. Associative property
3. Identity property
4. Inverse property
5. Commutative property

Types of algebraic structures on the basis of Properties of binary operation

1. Closure property = **Groupoid / Quasi-group**
2. Closure property + Associative property = **Semi-groups**
3. Closure property + Associative property + Identity property = **Monoids**
4. Closure property + Associative property + Identity property + Inverse property = **Groups**
5. Closure property + Associative property + Identity property + Inverse property + Commutative property = **Abelian Groups**

7 Marks

(1) **Groupoid / Quasi – Groups**

Let G be a non empty set and $*$ be a binary operation on G .

Then $(G, *)$ is said to be Groupoid if it satisfied the following property

(i) Closure property

$\frac{2}{3}$

Example

$(N, +)$ ✓	$(Z, +)$ ✓	$(R, +)$ ✓	$(Q, +)$ ✓
$(N, -)$ ✗	$(Z, -)$ ✓	$(R, -)$ ✓	$(Q, -)$ ✓
$(N, \times)$ ✓	$(Z, \times)$ ✓	$(R, \times)$ ✓	$(Q, \times)$ ✓
$(N, \div)$ ✗	$(Z, \div)$ ✗	$(R, \div)$ ✗	$(Q, \div)$ ✗

Q.1 Prove that the set of natural number N under operation 'multiplication' is a groupoid.

Given

$$N = \{1, 2, 3, 4, \dots\}$$

$(N, \cdot)$ is a groupoid if it satisfied

Closure property

We know that the multiplication of two natural number is also a natural number

$$\therefore a \cdot b \in N \quad \forall a, b \in N$$

$\Rightarrow N$ is closed under multiplication

(2) Semi groups

Let G be a non empty set and $*$ be a binary operation on G .

Then $(G, *)$ is said to be a semigroup if it satisfied following properties

(i) Closure property

(ii) Associative property

Examples

$(N, +)$ ✓	$(Z, +)$ ✓	$(R, +)$ ✓	$(Q, +)$ ✓
$(N, \times)$ ✓	$(Z, -)$ ✗	$(R, -)$ ✗	$(Q, -)$ ✗
	$(Z, \times)$ ✓	$(R, \times)$ ✓	$(Q, \times)$ ✓

Q.2 Prove that the set of natural number N under operation 'addition' is a semi group.

Given

$$N = \{1, 2, 3, 4, \dots\}$$

$(N, +)$ is a Semi-group if it satisfied

(i) closure property

We know that addition of two natural number is also a natural number

$$\therefore a + b \in N \quad \forall a, b \in N$$

$\Rightarrow N$ is closed under addition

(ii) Associative property

We know that addition of natural numbers is always associative

$$\therefore \forall a, b, c \in N$$

We have

$$(a + b) + c = a + (b + c)$$

Hence

$(N, +)$ is a semi-group

(3) Monoids

Let G be a non empty set and $*$ be a binary operation on G .

Then $(G, *)$ is said to be monoids if it satisfied the following properties

- (i) Closure property
- (ii) Associative property
- (iii) Identity property

Examples

$(N, +)$ ✗	$(Z, +)$ ✓	$(R, +)$ ✓	$(Q, +)$ ✓
$(N, \times)$ ✓	$(Z, \times)$ ✓	$(R, \times)$ ✓	$(Q, \times)$ ✓

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Q.3 Prove that $(N, \cdot)$ is a monoid where $\cdot$ is multiplication as a binary operation on the set of natural number

Given

$$N = \{1, 2, 3, 4, \dots\}$$

$(N, \cdot)$ is a monoid if it satisfied

(i) closure property

We know that the multiplication of two natural number is also a natural number

$$\therefore a \cdot b \in N \quad \forall a, b \in N$$

$\Rightarrow N$ is closed under multiplication

(ii) Associative property

We know that the multiplication of natural numbers is always associative

$$\therefore \forall a, b, c \in N$$

We have

$$(a \cdot b) \cdot c = a \cdot (b \cdot c)$$

(iii) Identity Property

$\forall a \in N \exists$ an element $1 \in N$ such that

$$a \cdot 1 = 1 \cdot a = a$$

$\Rightarrow 1$ is an identity element of N

Hence

$(N, \cdot)$ is a Monoid

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(4) Group

Let G be a non empty set and $*$ be a binary operation on G .

Then $(G, *)$ is said to be group if it satisfied the following properties

- (i) Closure property
- (ii) Associative property
- (iii) Identity property
- (iv) Inverse property

Examples

	$(\mathbb{Z}, +)$ ✓	$(\mathbb{R}, +)$ ✓	$(\mathbb{Q}, +)$ ✓
$(\mathbb{N}, \times)$ ✗	$(\mathbb{Z}, \times)$ ✗	$(\mathbb{R}, \times)$ ✗	$(\mathbb{Q}, \times)$ ✗

Q.4 Show that the set I of all integers is a group with respect to the operation of addition.

Given

$$I = \{ \dots -3, -2, -1, 0, 1, 2, 3 \dots \}$$

$(I, +)$ is a group if it satisfied

(i) closure property

We know that addition of two integer is also a integer

$$\therefore a + b \in I \quad \forall a, b \in I$$

$\Rightarrow I$ is closed under addition

(ii) Associative property

We know that addition of two integers is always associative

$$\therefore \forall a, b, c \in I$$

We have

$$(a + b) + c = a + (b + c)$$

(iii) Identity Property

$\forall a \in I \exists$ an element
 $0 \in I$ such that

$$a + 0 = a = 0 + a$$

$\Rightarrow 0$ is the identity element of I

(iv) Inverse property

$\forall a \in I \exists$ an element $(-a) \in I$
 such that

$$a + (-a) = (-a) + a = 0$$

$\Rightarrow$ Each element of I possesses
 additive inverse

(5) Abelian Group

Let G be a non empty set and $*$ be a binary operation on G .

Then $(G, *)$ is said to be Abelian group if it satisfied the following properties

- (i) Closure property
- (ii) Associative property
- (iii) Identity property
- (iv) Inverse property
- (v) Commutative property

	$(\mathbb{Z}, +)$ ✓	$(\mathbb{R}, +)$ ✓	$(\mathbb{Q}, +)$ ✓
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Q.5 Prove that the set all rational numbers Q form an Abelian group under addition.

Given

$$Q = \{x : x \text{ is a rational number}\}$$

$(Q, +)$ is an abelian group if it satisfied

(i) closure property

We know that addition of two rational number is also a rational number

$$\therefore a+b \in Q \quad \forall a, b \in Q$$

$\Rightarrow Q$ is closed under addition

(ii) Associative property

We know that addition of rational numbers is always associative

$$\therefore \forall a, b, c \in Q$$

We have

$$(a+b)+c = a+(b+c)$$

(iii) Identity Property

$\forall a \in Q \exists$ an element $0 \in Q$ such that

$$a + 0 = 0 + a = a$$

$\Rightarrow 0$ is additive identity of Q

(iv) Inverse property

$\forall a \in Q \exists$ an element $(-a) \in Q$ such that

$$a + (-a) = (-a) + a = 0$$

$\Rightarrow$ Each element possesses additive inverse

(v) commutative property

$$\forall a, b \in Q$$

We have

$$a + b = b + a$$

Hence,

$(Q, +)$ is an Abelian group

Topic : *Different types of algebraic structures*

Q.1 *Prove that the set of all real numbers R form an Abelian group under addition.*

Q.2 *Prove that the set all integer numbers N form an Abelian group under addition.*

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DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

DISCRETE MATHEMATICS

UNIT -4 : Algebraic Structures

Lecture - 03

Today's Target

- *Question based on infinite abelian group (PART – 1)*
- PYQ
- DPP

(1) Closer Property

G is closed under the operation $*$ if

$$\forall a, b \in G \quad \text{then } a * b \in G$$

(2) Associative Property

The operation $*$ on G is associative if

$$\forall a, b, c \in G$$

we have

$$(a * b) * c = a * (b * c)$$

(3) Identity Property (Existence of identity element)

The operation $*$ on G has an identity property if

$$\forall a \in G \quad \exists \text{ an identity element } e \in G \text{ such that}$$

$$a * e = a = e * a$$

(4) Inverse Property (*Existence of inverse element*)

The operation $*$ on G has an inverse property if

$\forall a \in G \quad \exists$ an inverse element $b \in G$ such that

$$a * b = b * a = e$$

(5) Commutative Property

The operation $*$ on G is commutative if

$\forall a, b \in G$

we have

$$a * b = b * a$$

Different types of algebraic structures

1. Closure property = **Groupoid / Quasi-group**

2. Closure property + Associative property = **Semi-groups**

3. Closure property + Associative property + Identity property = **Monoids**

4. Closure property + Associative property + Identity property + Inverse property = **Groups**

5. Closure property + Associative property + Identity property + Inverse property + Commutative property = **Abelian Groups**

Order of Group

The number of elements in a group is called the order of the group.

It is denoted by $O(G)$

Finite and Infinite Group

➤ A group $(G, *)$ is called an infinite group if G is an infinite set

$$O(G) = \infty$$

➤ A group $(G, *)$ is called a finite group if G is a finite set

$$O(G) = \text{Finite}$$

Question based on infinite abelian group

$$b * c = b + c + 1$$

Q.1 Define the binary operation $*$ on $\mathbb{Z}$ by $a * b = a + b + 1$ for all a, b belongs to set of integers.

Verify that $(\mathbb{Z}, *)$ is abelian group? Discuss the properties of abelian group.

(AKTU-2021-22)

Given

$$a * b = a + b + 1$$

(i) closure property

$$\forall a, b \in \mathbb{Z}$$

$$a + b \in \mathbb{Z}$$

$$a + b + 1 \in \mathbb{Z}$$

$$\Rightarrow a * b \in \mathbb{Z}$$

$\therefore$ Satisfied closure property

(ii) Associative property

$$\forall a, b, c \in \mathbb{Z}$$

$$(a * b) * c = \overbrace{(a + b + 1)}^a * \overbrace{c}^b$$

$$= (a + b + 1) + c + 1$$

$$= a + b + c + 2$$

$$a * (b * c) = a * (b + c + 1)$$

$$= a + (b + c + 1) + 1$$

$$= a + b + c + 2$$

$$\Rightarrow (a * b) * c = a * (b * c)$$

Satisfied Associative property

(III) Identity property

$\forall a \in \mathbb{Z} \exists$ an identity element $e \in \mathbb{Z}$ such that

$$a * e = a$$

$$a + e + 1 = a$$

$$e + 1 = 0$$

$$e = -1$$

$$\Rightarrow e \in \mathbb{Z}$$

$\therefore$ Satisfied identity property

(IV) Inverse property

$\forall a \in \mathbb{Z} \exists$ an inverse element $b \in \mathbb{Z}$ such that

$$a * b = e$$

$$a + b + 1 = -1$$

$$a + b = -2$$

$$b = (-2 - a) \in \mathbb{Z}$$

$$\Rightarrow b \in \mathbb{Z}$$

$\therefore$ Satisfied inverse property

(v) commutative property

$$\forall a, b \in \mathbb{Z}$$

$$a * b = a + b + 1$$

$$b * a = b + a + 1$$

$$\Rightarrow a * b = b * a$$

$\therefore$ Satisfied commutative property

Hence

$(\mathbb{Z}, *)$ is an abelian group

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Q.2 Consider an algebraic system $(Q, *)$ where Q is the set of rational numbers other than 1 and $*$ is a binary operation defined by $a * b = a + b - ab$ for all a, b belongs to set of rational numbers. Verify that $(Q, *)$ is abelian group? (UPTU-2014-15)

Given

$$a * b = a + b - ab$$

(i) closure property

$$\forall a, b \in Q$$

$$\Rightarrow a + b \in Q$$

$$\Rightarrow ab \in Q$$

$$\Rightarrow a + b - ab \in Q$$

$$\Rightarrow a * b \in Q$$

Satisfied closure property

(ii) Associative property

$$\forall a, b, c \in Q$$

$$(a * b) * c = \overbrace{(a + b - ab)}^a * \overbrace{c}^b$$

$$= (a + b - ab) + c - (a + b - ab)c$$

$$= a + b - ab + c - (ac + bc - abc)$$

$$= a + b + c - ab - ac - bc + abc$$

$$(a * b) * c = a + b + c - ab - bc - ac + abc$$

$$a * (b * c) = \overbrace{a}^a * \overbrace{(b + c - bc)}^b$$

$$= a + (b + c - bc) - a(b + c - bc)$$

$$= a + b + c - bc - ab - ac + abc$$

$$= a + b + c - ab - bc - ac + abc$$

$$\Rightarrow (a * b) * c = a * (b * c)$$

$\therefore$ Associative property is
Satisfied

(iii) Identity property

$\forall a \in \mathbb{Q} \exists$ an identity element

$e \in \mathbb{Q}$ such that

$$a * e = a$$

$$a + e - ae = a$$

$$e - ae = 0$$

$$e(1 - a) = 0$$

$$\therefore 1 - a \neq 0$$

$$\Rightarrow e = 0 \in \mathbb{Q}$$

Satisfied identity property

(IV) Inverse property

$\forall a \in \mathbb{Q} \exists$ an inverse element

$b \in \mathbb{Q}$ such that

$$a * b = e$$

$$a * b = 0$$

$$a + b - ab = 0$$

$$b(1-a) = -a$$

$$b = \frac{-a}{1-a} = \frac{+a}{+(a-1)}$$

$$\boxed{b = \frac{a}{a-1}} \in \mathbb{Q} \quad \left\{ \because a \neq 1 \right.$$

Satisfied inverse property

(V) commutative property

$$\forall a, b \in \mathbb{Q}$$

$$a * b = a + b - ab$$

$$b * a = b + a - ba = a + b - ab$$

$$\Rightarrow a * b = b * a$$

$\Rightarrow$ Satisfied commutative property
 $(\mathbb{Z}, *)$ is an abelian group

Q.3 Let G be the set of all non-zero real number and let $a * b = ab/2$.

Show that $(G, *)$ be an abelian group. (GBTU-2011)

Given

$$a * b = \frac{ab}{2}$$

(i) closure property

$$\forall a, b \in G$$

$$\Rightarrow ab \in G$$

$$\Rightarrow \frac{ab}{2} \in G$$

$$\Rightarrow a * b \in G$$

$\therefore$ satisfied closure property

(ii) Associative property

$$\forall a, b, c \in G$$

$$(a * b) * c = \underbrace{\left(\frac{ab}{2}\right)}_a * \underbrace{c}_b$$

$$= \frac{\frac{ab}{2} \times c}{2}$$

$$= \frac{abc}{4}$$

$$a * (b * c) = a * \left(\frac{bc}{2} \right)$$

$$= \frac{a \frac{bc}{2}}{2}$$

$$= \frac{abc}{4}$$

$$\Rightarrow (a * b) * c = a * (b * c)$$

$\therefore$ Satisfied Associative property

(iii) Identity property

$\forall a \in G \exists$ an identity element $e \in G$

such that

$$a * e = a$$

$$\frac{ae}{2} = a$$

$$ae = 2a$$

$$ae - 2a = 0$$

$$a(e - 2) = 0$$

$$\therefore a \neq 0$$

$$\therefore e - 2 = 0$$

$$e = 2 \in G$$

$$e \in G$$

$\therefore$ Satisfied identity property

(IV) Inverse property

$\forall a \in G \exists$ an inverse element

$b \in G$ such that

$$a * b = e$$

$$a * b = 2$$

$$\frac{ab}{2} = 2$$

$$b = \frac{4}{a} \in G$$

$$b \in G$$

$\therefore$ satisfied inverse property

(V) commutative property

$\forall a, b \in G$

$$a * b = \frac{ab}{2}$$

$$b * a = \frac{ba}{2} = \frac{ab}{2}$$

$$\Rightarrow a * b = b * a$$

$\therefore$ Satisfied commutative property

Hence

$(G, *)$ is an abelian group

Topic : Question based on infinite abelian group

Q.1 Prove that the set of all non zero real number is an abelian group w.r.t. to multiplication as binary operation (GBTU-2009)

Q.2 Let G be a group of integers with binary operation $*$ defined by

$$a * b = a + b - 2 \quad \forall \quad a, b \in G$$

Find the identity element of the group $(G, *)$

(UPTU-2014-15)

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

DISCRETE MATHEMATICS

UNIT -4 : Algebraic Structures

Lecture - 04

Today's Target

- *Question based on infinite abelian group (Part – 2)*
- PYQ
- DPP

(1) Closer Property

G is closed under the operation $*$ if

$$\forall a, b \in G \quad \text{then } a * b \in G$$

(2) Associative Property

The operation $*$ on G is associative if

$$\forall a, b, c \in G$$

we have

$$(a * b) * c = a * (b * c)$$

(3) Identity Property (Existence of identity element)

The operation $*$ on G has an identity property if

$$\forall a \in G \quad \exists \text{ an identity element } e \in G \text{ such that}$$

$$a * e = a = e * a$$

(4) **Inverse Property** (Existence of inverse element)

The operation $*$ on G has an inverse property if

$\forall a \in G \quad \exists$ an inverse element $b \in G$ such that

$$a * b = b * a = e$$

(5) **Commutative Property**

The operation $*$ on G is commutative if

$\forall a, b \in G$

we have

$$a * b = b * a$$

Q.1 Let Q be the set of positive rational numbers which can be expressed in the form $2^a 3^b$, where a and b are integers. Prove that the algebraic structure $(Q, \cdot)$ is a group where $\cdot$ is a multiplication operator.

(AKTU-2020)

Given

$$Q = \{2^a 3^b : a, b \in \mathbb{Z}\}$$

(i) closure property

$$\text{Let } q_1 = 2^a 3^b, q_2 = 2^c 3^d$$

$$\begin{aligned} q_1 \cdot q_2 &= 2^a 3^b \cdot 2^c 3^d \\ &= 2^{a+c} \cdot 3^{b+d} \end{aligned}$$

Where $a, b, c, d \in \mathbb{Z}$

$$q_1 \cdot q_2 = 2^{a+c} \cdot 3^{b+d}$$

$$a+c, b+d \in \mathbb{Z}$$

$$q_1 \cdot q_2 = 2^{a+c} \cdot 3^{b+d} \in Q$$

Satisfied closure property

(ii) Associative property

$$\text{Let } q_1 = 2^a 3^b, q_2 = 2^c 3^d, q_3 = 2^e 3^f$$

$$(q_1 \cdot q_2) \cdot q_3 = (2^a 3^b \cdot 2^c 3^d) \cdot 2^e 3^f$$

$$= 2^{a+c} 3^{b+d} \cdot 2^e 3^f$$

$$= 2^{(a+c)+e} 3^{(b+d)+f}$$

$$q_1 \cdot (q_2 \cdot q_3) = 2^a 3^b \cdot (2^c 3^d \cdot 2^e 3^f)$$

$$= 2^a 3^b \cdot (2^{c+e} 3^{d+f})$$

$$= 2^{a+c+e} 3^{b+d+f}$$

$$\Rightarrow (q_1 \cdot q_2) \cdot q_3 = q_1 \cdot (q_2 \cdot q_3)$$

$\therefore$ Satisfied Associative property

(iii) Identity Property

$\forall 2^a 3^b \in Q \exists$ an identity element

$e \in Q$ such that

$$e \cdot 2^a 3^b = 2^a 3^b$$

$$e = \frac{2^a 3^b}{2^a 3^b}$$

$$e = 2^0 3^0 = 1 \in Q$$

Satisfied Identity property

(IV) Inverse property

$\forall 2^a 3^b \in Q \exists$ an inverse element

$y \in Q$ such that

$$y \cdot 2^a 3^b = e$$

$$y \cdot 2^a 3^b = 1$$

$$y = \frac{1}{2^a 3^b}$$

$$y = 2^{-a} 3^{-b} \in Q$$

Satisfied inverse property

Hence

$(Q, \cdot)$ is a group

Q.2 Show that the set $G = \{x + y\sqrt{2} \mid x, y \in \mathbb{Q}\}$ is a group with respect to Multiplication.

Given

$$G = \{x + y\sqrt{2} \mid x, y \in \mathbb{Q}\}$$

(i) Closure property

$$\text{Let } X = x_1 + y_1\sqrt{2}, Y = x_2 + y_2\sqrt{2} \in G$$

$$\text{Where } x_1, y_1, x_2, y_2 \in \mathbb{Q}$$

$$XY = (x_1 + y_1\sqrt{2}) \cdot (x_2 + y_2\sqrt{2})$$

$$= x_1x_2 + x_1y_2\sqrt{2} + y_1x_2\sqrt{2} + 2y_1y_2$$

$$= x_1x_2 + 2y_1y_2 + \sqrt{2}(x_1y_2 + y_1x_2)$$

$$XY = (x_1x_2 + 2y_1y_2) + \sqrt{2}(x_1y_2 + y_1x_2)$$

$$\therefore (x_1x_2 + 2y_1y_2) \in \mathbb{Q}$$

$$(x_1y_2 + y_1x_2) \in \mathbb{Q}$$

$$\Rightarrow XY \in G$$

Satisfied closure law

(II) Associative property

Let $X = x_1 + y_1\sqrt{2}$, $Y = x_2 + y_2\sqrt{2}$, $Z = x_3 + y_3\sqrt{2} \in G$

where $x_1, y_1, x_2, y_2, x_3, y_3 \in \mathbb{Q}$

$$(X \cdot Y) \cdot Z = [(x_1 + y_1\sqrt{2}) \cdot (x_2 + y_2\sqrt{2})] \cdot (x_3 + y_3\sqrt{2})$$

$$= (x_1 + y_1\sqrt{2}) \cdot ((x_2 + y_2\sqrt{2}) \cdot (x_3 + y_3\sqrt{2}))$$

$$= (x_1 + y_1\sqrt{2}) \cdot [(x_2 + y_2\sqrt{2}) \cdot (x_3 + y_3\sqrt{2})]$$

$\therefore$ Multiplication of Rational numbers
association

$$(X \cdot Y) \cdot Z = X \cdot (Y \cdot Z)$$

satisfied associative
property.

(III) Identity property

$\forall (x + y\sqrt{2}) \exists$ an identity
element $e \in G$ such that

$$e \cdot (x + y\sqrt{2}) = (x + y\sqrt{2})$$

$$e = 1$$

$$e = 1$$

$$e = 1 + 0\sqrt{2} \in G$$

Satisfied identity property

(iv) Inverse property

$\forall (n+y\sqrt{2}) \in G \exists$ an inverse element a such that

$$a \cdot (n+y\sqrt{2}) = e$$

$$a \cdot (n+y\sqrt{2}) = 1$$

$$a = \frac{1}{n+y\sqrt{2}}$$

$$a = \frac{1}{(n+y\sqrt{2})} \times \frac{n-y\sqrt{2}}{n-y\sqrt{2}}$$

$$a = \frac{n-y\sqrt{2}}{(n^2-2y^2)}, \quad n^2-2y^2 \neq 0$$

$$a = \frac{n}{n^2-2y^2} - \frac{y\sqrt{2}}{n^2-2y^2} \in G$$

$\therefore$ Satisfied inverse property

Hence, G is a group

Q.3 Let $G = \{ (a, b) : a, b \in \mathbb{R}, a \neq 0 \}$. Define a binary operation $*$ on G by

$(a, b) * (c, d) = (ac, bc + d) \quad \forall (a, b) * (c, d) \in G$. Show that $(G, *)$ is a group but not an abelian group

Given

$$G = \{ (a, b) : a, b \in \mathbb{R}, a \neq 0 \}$$

$$(a, b) * (c, d) = (ac, bc + d)$$

(i) Closure property

$$\text{Let } (a, b), (c, d) \in G$$

$$\text{Where } a, b, c, d \in \mathbb{R}$$

$$\text{and } a \neq 0, c \neq 0$$

$$(a, b) * (c, d) = (ac, bc + d) \quad (\text{AKTU-2020})$$

$$\therefore ac \in \mathbb{R} \text{ and}$$

$$bc + d \in \mathbb{R}$$

$$ac \neq 0$$

$$\Rightarrow (a, b) * (c, d) \in G$$

$\therefore$ Satisfied closure property

(ii) Associative property

Let $(a, b), (c, d), (e, f) \in G$

Where $a, b, c, d, e, f \in R$ and

$$a \neq 0, c \neq 0, e \neq 0$$

$$\begin{aligned} [(a, b) * (c, d)] * (e, f) &= (ac, bc + d) * (e, f) \\ &= (ace, bce + de + f) \end{aligned}$$

$$\begin{aligned} (a, b) * [(c, d) * (e, f)] &= (a, b) * (ce, de + f) \\ &= (ace, bce + de + f) \end{aligned}$$

$$[(a, b) * (c, d)] * (e, f) = (a, b) * [(c, d) * (e, f)]$$

$\therefore$ satisfied associative property

(iii) Identity Property

$\forall (a, b) \in G \exists$ an identity element $(n, y) \in G$ such that

$$(n, y) * (a, b) = (a, b)$$

$$(an, ay + b) = (a, b)$$

$$(an, ay+b) = (a, b)$$

$$\begin{array}{l|l} an = a & ay+b = b \\ n = 1 & ay = 0 \\ & y = 0 \end{array}$$

$$(n, y) = (1, 0) \in G$$

$\Rightarrow$ satisfied identity property

(iv) Inverse property

$\forall (a, b) \in G \exists$ an inverse element

$(n, y) \in G$ such that

$$(n, y) * (a, b) = e$$

$$(n, y) * (a, b) = (1, 0)$$

$$(an, ay+b) = (1, 0)$$

$$\begin{array}{l|l} an = 1 & ay+b = 0 \\ n = \frac{1}{a} & y = -\frac{b}{a} \end{array}$$

$$(x, y) = \left(\frac{1}{a}, -\frac{b}{a}\right) \in G$$

$\therefore$ satisfied inverse property

Hence, $(G, *)$ is a group

(iii) Commutative property

$$\text{Let } (a, b), (c, d) \in G$$

Where $a, b, c, d \in \mathbb{R}$ and

$$a \neq 0, c \neq 0$$

$$(a, b) * (c, d) = (ac, bc + d)$$

$$(c, d) * (a, b) = (ca, da + b)$$

$$\Rightarrow (a, b) * (c, d) \neq (c, d) * (a, b)$$

$\Rightarrow$ Not satisfied commutative property

Hence

$(G, *)$ is a group but not a abelian group

Topic : Question based on infinite abelian group (Part – 2)

Q.1 Show that the set $G = \{x + y\sqrt{2} \mid x, y \in \mathbb{Q}\}$ is a group with respect to addition. (UPTU-2014-15)

Q.2 Show that the set of matrices $A_\alpha = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}, \alpha \in \mathbb{R}$ forms abelian group under matrix multiplication.

Given

$$G = \left\{ A_\alpha = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}, \alpha \in \mathbb{R} \right\}$$

A_α, A_β

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

DISCRETE MATHEMATICS

UNIT -4 : Algebraic Structures

Lecture - 05

Today's Target

- *Questions based on Finite abelian group*
- PYQ
- DPP

(1) Closer Property

G is closed under the operation $*$ if

$$\forall a, b \in G \quad \text{then } a * b \in G$$

(2) Associative Property

The operation $*$ on G is associative if

$$\forall a, b, c \in G$$

we have

$$(a * b) * c = a * (b * c)$$

(3) Identity Property (Existence of identity element)

The operation $*$ on G has an identity property if

$$\forall a \in G \quad \exists \text{ an identity element } e \in G \text{ such that}$$

$$a * e = a = e * a$$

(4) Inverse Property (*Existence of inverse element*)

The operation $*$ on G has an inverse property if

$\forall a \in G \quad \exists$ an inverse element $b \in G$ such that

$$a * b = b * a = e$$

(5) Commutative Property

The operation $*$ on G is commutative if

$\forall a, b \in G$
we have

$$a * b = b * a$$

Q.1 Show that the Set of cube root of unity is an abelian group with respect to Multiplication.

$$\text{Let } \omega = 1^{\frac{1}{3}}$$

$$\omega^3 = 1$$

$$\omega^3 - 1 = 0$$

$$\omega^3 - 1^3 = 0$$

$$(\omega - 1)(\omega^2 + \omega + 1) = 0$$

$$\omega - 1 = 0$$

$$\boxed{\omega = 1}$$

$$\omega^2 + \omega + 1 = 0$$

$$\omega = \frac{-1 \pm \sqrt{1-4}}{2}$$

$$\omega = \frac{-1 \pm \sqrt{-3}}{2}$$

$$\omega = \frac{-1 \pm i\sqrt{3}}{2}$$

$$\boxed{\omega = \frac{-1}{2} + i\frac{\sqrt{3}}{2}} = \omega$$

$$\boxed{\omega = \frac{-1}{2} - i\frac{\sqrt{3}}{2}} = \omega^2$$

Hence

(GBTU-2013)

$$\omega = 1, \omega, \omega^2$$

Note - $\omega^3 = 1$

Let $G = \{1, \omega, \omega^2\}$

composition table

$\cdot$	1	ω	ω^2
1	1	ω	ω^2
ω	ω	ω^2	1
ω^2	ω^2	1	ω

(i) Closure Property: Since all the entries in the composition table are the elements of the set G and hence closure property is satisfied

$$\forall a, b \in G$$

$$a \cdot b \in G$$

(ii) Associative Property: The elements of the set G are all complex numbers and multiplication of complex numbers obey's associative law

$$\forall a, b, c \in G$$

we have

$$(a \cdot b) \cdot c = a \cdot (b \cdot c)$$

(iii) **Identity Property:** It can be seen from the first row and first column of the composition table.

$\forall a \in G \exists$ an identity element $1 \in G$

Such that

$$a \cdot 1 = 1 \cdot a = a$$

Identity element = 1

✓ (iv) **Inverse Property:** From the composition table it is clear that

$$1 \cdot 1 = 1$$

$$w \cdot w^2 = w^3 = 1$$

$$w^2 \cdot w = w^3 = 1$$

$$1^{-1} = 1$$

$$w^{-1} = w^2$$

$$(w^2)^{-1} = w$$

Hence

$\forall a \in G \exists$ an inverse element b such that

$$a \cdot b = 1$$

(v) **Commutative Property:** From the composition table it is clear that elements in each row and corresponding ^{column} are equal.

$$\forall a, b \in G$$

we have

$$a \cdot b = b \cdot a$$

So, satisfied commutative property

$(G, \cdot)$ is a finite abelian group of order 3

Q.2 Show that the set of four fourth roots of unity $1, -1, +i, -i$ forms an abelian group with respect to Multiplication.

(AKTU 2022-2023)

Let $x = 1^{1/4}$

$$x^4 = 1$$

$$x^4 - 1 = 0$$

$$(x^2 - 1)(x^2 + 1) = 0$$

$$(x - 1)(x + 1)(x^2 + 1) = 0$$

$$x - 1 = 0$$

$$x = 1$$

$$x + 1 = 0$$

$$x = -1$$

$$x^2 + 1 = 0$$

$$x^2 = -1$$

$$x = \pm \sqrt{-1}$$

$$x = \pm i$$

$$x = i$$

$$x = -i$$

$$x = 1, -1, i, -i$$

Note

$$i^2 = -1$$

$$i^3 = -i$$

$$i^4 = 1$$

$$G = \{1, -1, i, -i\}$$

Composition table

$\cdot$	1	-1	i	-i
1	1	-1	i	-i
-1	-1	1	-i	i
i	i	-i	-1	1
-i	-i	i	1	-1

(i) Closure Property: Since all the entries in the composition table are the elements of the set G and hence closure property is satisfied

$$\forall a, b \in G$$

$$a \cdot b \in G$$

(ii) Associative Property: The elements of the set G are all complex numbers and multiplication of complex numbers obey's associative law

$$\forall a, b, c \in G$$

we have

$$(a \cdot b) \cdot c = a \cdot (b \cdot c)$$

(iii) Identity Property: It can be seen from the first row and first column of the composition table $\forall a \in G \exists$ an identity element $1 \in G$

such that

$$a \cdot 1 = 1 \cdot a = a$$

So, identity property is satisfied

(iv) Inverse Property: From the composition table it is clear that

$$\begin{aligned} 1 \cdot 1 &= 1 \\ (-1) \cdot (-1) &= 1 \\ i \cdot (-i) &= 1 \\ (-i) \cdot (i) &= 1 \end{aligned}$$

i.e

$$\begin{aligned} 1^{-1} &= 1 \\ -1^{-1} &= -1 \\ i^{-1} &= -i \\ -i^{-1} &= i \end{aligned}$$

$\forall a \in G \exists$ an inverse element $b \in G$ such that

$$a \cdot b = 1$$

So, satisfied inverse property

(v) **Commutative Property:** From the composition table it is clear that elements in each row and corresponding are equal.

$$\forall a, b \in G$$

$$a \cdot b = b \cdot a$$

So, satisfied commutative property

Hence

$(G, \cdot)$ is a finite abelian group of order 4

Q.3 Show that the Set of n n th roots of unity form a finite abelian group of order n with respect to Multiplication.

$$1^{1/n} = (1 + 0i)^{1/n}$$

$$= (\cos 0 + i \sin 0)^{1/n}$$

$$= (\cos 2r\pi + i \sin 2r\pi)^{1/n}$$

By De-Moivre's theorem

$$= \cos \frac{2r\pi}{n} + i \sin \frac{2r\pi}{n}$$

By $e^{i\theta} = \cos \theta + i \sin \theta$

$$1^{1/n} = e^{i \frac{2r\pi}{n}} = \left(e^{i \frac{2\pi}{n}} \right)^r$$

(GBTU-2013)

$$1^{1/n} = \omega^r$$

where $r = 0, 1, 2, 3, \dots, n-1$

$$1^{1/n} = 1, \omega, \omega^2, \omega^3, \dots, \omega^{n-1}$$

$$G = \{ 1, \omega, \omega^2, \omega^3, \dots, \omega^{n-1} \}$$

(i) Closure Property:

$$1 \cdot \omega = \omega$$

$$\omega \cdot \omega^2 = \omega^3 = 1$$

$$\omega^2 \cdot \omega^3 = \omega^5 = \omega^3 \omega^2 = \omega^2$$

$$\omega^5 \cdot \omega^7 = \omega^{12} = (\omega^3)^4 = 1$$

$$\omega^9 \cdot \omega^{11} = \omega^{20} = \omega^2$$

and so on

$$\forall a, b \in G$$

$$a \cdot b \in G$$

So satisfied closure property

(ii) Associative Property: The elements of the set G are all complex numbers and multiplication of complex numbers obey's associative law

$$\forall a, b, c \in G$$

we have

$$(a \cdot b) \cdot c = a \cdot (b \cdot c)$$

(iii) Identity Property:

$$1 \cdot \omega = \omega$$

$$1 \cdot \omega^2 = \omega^2$$

$$1 \cdot \omega^3 = \omega^3$$

and so on

$\forall a \in G \exists$ an identity element

$1 \in G$ such that

$$a \cdot 1 = 1 \cdot a = a$$

(iv) Inverse Property:

$$1 \cdot 1 = 1$$

$$\omega \cdot \omega^2 = 1$$

$$\omega^2 \cdot \omega = 1$$

$$\omega^3 \cdot 1 = 1$$

$$\omega^4 \cdot \omega^2 = \omega^6 = 1$$

and so on

$\forall a \in G \exists$ an inverse element

$b \in G$ such that

$$a \cdot b = b \cdot a$$

(v) Commutative Property:

We know that multiplication of complex number is commutative

$$\forall a, b \in G$$

we have

$$a \cdot b = b \cdot a$$

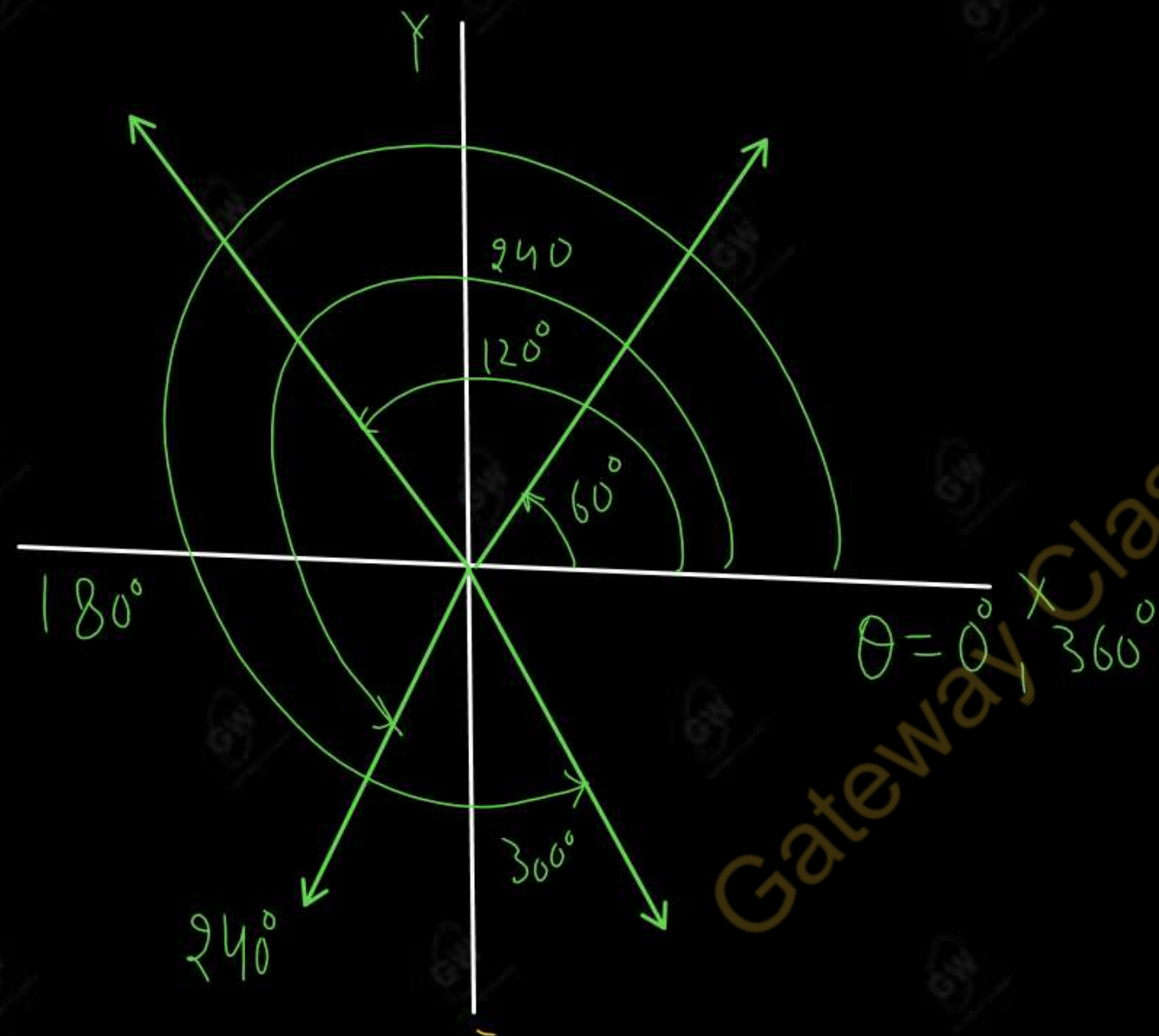
So, satisfied commutative property

Hence

$(G, \cdot)$ is a finite abelian group of order n

Q. 4 Let $R = \{ 0^\circ, 60^\circ, 120^\circ, 180^\circ, 240^\circ, 300^\circ \}$ and $*$ is a binary operation, $a*b$ is overall angular rotation. Show that $(R, *)$ is a group

$$R = \{ 0^\circ, 60^\circ, 120^\circ, 180^\circ, 240^\circ, 300^\circ \}$$



Composition table

+	0°	60°	120°	180°	240°	300°
0°	0°	60°	120°	180°	240°	300°
60°	60°	120°	180°	240°	300°	0°
120°	120°	180°	240°	300°	0°	60°
180°	180°	240°	300°	0°	60°	120°
240°	240°	300°	0°	60°	120°	180°
300°	300°	0°	60°	120°	180°	240°

Topic : Question based on Finite abelian group

Q.1 Let $R = \{ 0^\circ, 60^\circ, 120^\circ, 180^\circ, 240^\circ, 300^\circ \}$ and $*$ is a binary operation, $a*b$ is overall angular rotation. Show that $(R, *)$ is a group

Q.2 Show that the matrices

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad D = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Form a multiplicative abelian group.

composition table

$\cdot$	A	B	C	D
A	A	B	C	D
B	B	B	C	D
C	C	C	C	D
D	D	D	D	D

$$G = \{ A, B, C, D \}$$

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

DISCRETE MATHEMATICS

UNIT -4 : Algebraic Structures

Lecture - 06

Today's Target

- ***Addition and Multiplication Modulo***
- PYQ
- DPP

Addition Modulo m ($+_m$)

Let a and b be any two integers and m be a positive integer greater than 1.

The addition modulo m of a and b is defined as the least non-negative remainder r obtained when $a + b$ is divided by m . It is written as $a +_m b = r$

Where $0 \leq r < m$

Examples

(i) $5 +_3 8 = 1$

(ii) $15 +_5 35 = 0$

(iii) $5 +_4 5 = 2$

(iv) $3 +_5 8 = 1$

(v) $16 +_7 10 = 5$

$$\begin{array}{r} 3 \overline{) 13} \quad (4 \\ \underline{12} \\ 1 \rightarrow R \end{array}$$

Multiplication Modulo m ($\times_m$)

Let a and b be any two integers and m be a positive integer greater than 1.

The Multiplication modulo m of a and b is defined as the least non-negative remainder r obtained when product ab is divided by m . It is written as $a \times_m b = r$

Where $0 \leq r < m$

Examples

(i) $5 \times_4 5 = 1$

(ii) $3 \times_4 5 = 3$

(iii) $9 \times_6 4 = 0$

(iv) $3 \times_5 8 = 4$

(v) $8 \times_7 5 = 5$

$$\begin{array}{r} 4 \overline{) 25} \\ \underline{24} \\ 1 \rightarrow R \end{array}$$

Gateway Classes: 7455 9672 234

Q.1 Show that $(G, +_8)$ is an abelian group Where $G = \{0, 1, 2, 3, 4, 5, 6, 7\}$.

OR

Prove that the set $Z_8 = \{0, 1, 2, 3, 4, 5, 6, 7\}$ is a finite abelian group of order 8 with respect to addition modulo 8

Form composition table

$+_8$	0	1	2	3	4	5	6	7
0	0	1	2	3	4	5	6	7
1	1	2	3	4	5	6	7	0
2	2	3	4	5	6	7	0	1
3	3	4	5	6	7	0	1	2
4	4	5	6	7	0	1	2	3
5	5	6	7	0	1	2	3	4
6	6	7	0	1	2	3	4	5
7	7	0	1	2	3	4	5	6

(i) **Closure Property:**

All the elements in the composition table are the elements of the set G

i.e $\forall a, b \in G$

$a +_8 b \in G$

$\therefore$ Satisfied closure property

(ii) Associative Property:

$$\forall a, b, c \in G$$

we have

$$(a +_8 b) +_8 c = a +_8 (b +_8 c)$$

Example

$$(2 +_8 4) +_8 7 = 2 +_8 (4 +_8 7)$$

$$6 +_8 7 = 2 +_8 3$$

$$\boxed{5 = 5}$$

$\therefore$ Satisfied Associative property

(iii) Identity Property:

It can be seen from the first row and first column of the composition table.

$\forall a \in G \exists$ an identity element $0 \in G$ such that

$$0 +_8 a = a = a +_8 0$$

$\therefore 0$ is the identity element

Hence

Satisfied identity property

(iv) Inverse Property:

$\forall a \in G \exists$ an inverse element $b \in G$ such that

$$a +_8 b = 0$$

From composition table

$$0 +_8 0 = 0 \Rightarrow 0^{-1} = 0$$

$$1 +_8 7 = 0 \Rightarrow 1^{-1} = 7$$

$$2 +_8 6 = 0 \Rightarrow 2^{-1} = 6$$

$$3 +_8 5 = 0 \Rightarrow 3^{-1} = 5$$

$$4 +_8 4 = 0 \Rightarrow 4^{-1} = 4$$

$$5 +_8 3 = 0 \Rightarrow 5^{-1} = 3$$

$$6 +_8 2 = 0 \Rightarrow 6^{-1} = 2$$

$$7 +_8 1 = 0 \Rightarrow 7^{-1} = 1$$

$\therefore$ Satisfied inverse property under $+_8$

(v) Commutative Property:

From the composition table it is clear that elements in each row and corresponding are equal.

i.e. $\forall a, b \in G$

we have

$$a +_8 b = b +_8 a$$

$\therefore$ satisfied commutative property

Hence

$(G, +_8)$ is a Finite abelian group of order 8

Q.2: Show that (G, X_7) is an abelian group Where $G = \{1, 2, 3, 4, 5, 6\}$.

OR

Prove that the set $G = \{1, 2, 3, 4, 5, 6\}$ is a finite abelian group of order 6 under the multiplication modulo 7

Form composition table

X_7	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	4	6	1	3	5
3	3	6	2	5	1	4
4	4	1	5	2	6	3
5	5	3	1	6	4	2
6	6	5	4	3	2	1

(i) Closure Property:

All the elements in the composition table are the elements of the set G

i.e $\forall a, b \in G$

$a X_7 b \in G$

$\therefore$ satisfied closure property
under X_7

(ii) Associative Property:

$$\forall a, b, c \in G$$

we have

$$(a \times b) \times c = a \times (b \times c)$$

Example

$$(3 \times 5) \times 6 = 3 \times (5 \times 6)$$

$$1 \times 6 = 3 \times 2$$

$$6 = 6$$

$\therefore$ satisfied associative property

(iii) Identity Property:

It can be seen from the first row and first column of the composition table.

$\forall a \in G \exists$ an identity element $1 \in G$ such that

$$a \times 1 = a = 1 \times a$$

$\therefore 1$ is the identity element

Hence

Satisfied identity property

(iv) Inverse Property:

$\forall a \in G \exists$ an inverse element $b \in G$ such that

$$a \times_7 b = 1$$

From composition table

$$1 \times_7 1 = 1 \Rightarrow 1^{-1} = 1$$

$$2 \times_7 4 = 1 \Rightarrow 2^{-1} = 4$$

$$3 \times_7 5 = 1 \Rightarrow 3^{-1} = 5$$

$$4 \times_7 2 = 1 \Rightarrow 4^{-1} = 2$$

$$5 \times_7 3 = 1 \Rightarrow 5^{-1} = 3$$

$$6 \times_7 6 = 1 \Rightarrow 6^{-1} = 6$$

$\therefore$ satisfied inverse property under $\times_7$

(v) Commutative Property:

From the composition table it is clear that elements in each row and corresponding are equal.

$$\text{i.e. } \forall a, b \in G$$

We have

$$a \times_7 b = b \times_7 a$$

$\Rightarrow$ satisfied commutative property under $\times_7$

Hence

$(G, \times_7)$ is a Finite abelian group of order 6

Q.3 Show that the set $\{1, 2, 3, 4, 5\}$ is not a group under addition and multiplication modulo 6.

$$\text{Let } G = \{1, 2, 3, 4, 5\}$$

Form composition table for addition modulo 6

$+_6$	1	2	3	4	5
1	2	3	4	5	0
2	3	4	5	0	1
3	4	5	0	1	2
4	5	0	1	2	3
5	0	1	2	3	4

Since all the elements of composition tables does not belongs to the set G

$$\text{As } 0 \notin G$$

$\therefore$ Not satisfied closure property

$\Rightarrow (G, +_6)$ is not a group

Form composition table for multiplication modulo 6

X_6	1	2	3	4	5
1	1	2	3	4	5
2	2	4	0	2	4
3	3	0	3	0	3
4	4	2	0	4	2
5	1	4	3	2	1

Since all the elements of the composition table does not belongs to the set G

As $0 \notin G$

$\therefore$ Not satisfied closure property

$\Rightarrow (G, \times_7)$ is not a group

Topic : Addition and Multiplication Modulo

Q.1 Prove that the set $Z_6 = \{0, 1, 2, 3, 4, 5\}$ is a finite abelian group of order 6 with respect to addition modulo 6

Q.2 Prove that the set $G = \{1, 2, 3, 4\}$ is an abelian group under multiplication modulo 5.

Gateway Classes : 7455 9612 84

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

DISCRETE MATHEMATICS

UNIT -4 : Algebraic Structures

Lecture - 07

Today's Target

- Order of an element of a group
- Some important Theorems of Group
- DPP
- PYQ

➤ The order of an element 'a' in a group G is the smallest positive integer 'n' such that

$$a^n = e$$

➤ The order of an element a is denoted by $O(a)$

$$O(a) = n$$

➤ If no such integer exists, then

$$O(a) = \infty$$

Note : (i) $O(e) = 1$

(ii) $O(a) = O(a^{-1})$

(iii) If $a^m = e \Rightarrow O(a) \leq m$

$$a^m = e$$

Example

$$(i)^4 = 1$$

$$(i)^8 = 1$$

$$(i)^{12} = 1$$

$$\omega^3 = 1$$

$$\omega^6 = 1$$

$$\omega^9 = 1$$

-- }

Q.1 Let $G = \{1, -1, i, -i\}$ be a multiplicative group.

Find the order of each element in G .

AKTU 2022 - 23

$$G = \{1, -1, i, -i\}$$

$$O(1) = 1$$

$\left\{ \because 1 \text{ is the identity element} \right\}$

$$(-1)^2 = 1$$

$$\Rightarrow O(-1) = 2$$

$$(i)^4 = i \times i \times i \times i = 1$$

$$\Rightarrow O(i) = 4$$

$$(-i)^4 = (-i) \times (-i) \times (-i) \times (-i) = 1$$

$$\Rightarrow O(-i) = 4$$

Q.2 Let $G = \{1, \omega, \omega^2\}$ be a multiplicative group. Find the order of every element.

$$G = \{1, \omega, \omega^2\}$$

$$O(1) = 1$$

$\left\{ \because 1 \text{ is the identity element} \right\}$

$$\omega \times \omega \times \omega = \omega^3 = 1$$

$$O(\omega) = 3$$

$$(\omega^2)^3 = \omega^6 = 1$$

$$O(\omega^2) = 3$$

Q.3 Find the order of each element of the group $G = \{0, 1, 2, 3, 4, 5, 6\}$, the composition in G is 'addition modulo 7'

$$G = \{0, 1, 2, 3, 4, 5, 6\}$$

$$O(0) = 1 \{ \because 0 \text{ is the identity element} \}$$

$$1 +_7 1 +_7 1 +_7 1 +_7 1 +_7 1 +_7 1 = 0$$

$$\Rightarrow O(1) = 7$$

$$2 +_7 2 +_7 2 +_7 2 +_7 2 +_7 2 +_7 2 = 0$$

$$\Rightarrow O(2) = 7$$

$$3 +_7 3 +_7 3 +_7 3 +_7 3 +_7 3 +_7 3 = 0$$

$$\Rightarrow O(3) = 7$$

Q4 Find the order of each element of $(R, +)$

$$R = \{x : x \text{ is a Real number}\}$$

$$o(0) = 1$$

$$\forall a \in R$$

$$o(a) = \infty \text{ \{except 0\}}$$

Since we cannot find

smallest integer

Q5 Find the order of every element in the

multiplication group $G = \{a, a^2, a^3, a^4, a^5, a^6 = e\}$

$$G = \{a, a^2, a^3, a^4, a^5, a^6\}$$

$$a^6 = e \text{ (Given)}$$

$$(a)^6 = a^6 = e \Rightarrow o(a) = 6$$

$$(a^2)^3 = a^6 = e \Rightarrow o(a^2) = 3$$

$$(a^3)^2 = a^6 = e \Rightarrow o(a^3) = 2$$

$$(a^4)^3 = a^{12} = e \Rightarrow o(a^4) = 3$$

$$(a^5)^6 = a^{30} = e \Rightarrow o(a^5) = 6$$

$$(a^6)^1 = a^6 = e \Rightarrow o(a^6) = 1$$

Note : The symbol $*$ is used for the group composition but we can write $a * b$ simply as ab in all theorems

Theorem-1 (Cancellation laws) : If a, b, c are three elements of a group G , then

$$ab = ac \Rightarrow b = c \quad (\text{Left cancellation law})$$

$$ba = ca \Rightarrow b = c \quad (\text{Right cancellation law})$$

Given

$$a, b, c \in G$$

Left cancellation law

$$ab = ac \quad (\text{Given})$$

$$a^{-1}(ab) = a^{-1}(ac) \quad \left\{ \text{Pre-multiplying by } a^{-1} \text{ on both side} \right\}$$

$$(a^{-1}a)b = (a^{-1}a)c \quad \left\{ \text{By Associative law} \right\}$$

$$eb = ec \quad \left\{ \because a^{-1}a = e \right\}$$

$$\boxed{b = c} \quad \left\{ \because ea = a \right\}$$

Hence proved

Right Cancellation law

$$ba = ca$$

$$(ba)a^{-1} = (ca)a^{-1} \left\{ \because \text{Post-Multiplying by } a^{-1} \text{ on both side} \right\}$$

$$b(aa^{-1}) = c(aa^{-1}) \left\{ \text{By Associative law} \right\}$$

$$be = ce$$

$$\boxed{b = c}$$

Hence proved

$$\left\{ \because aa^{-1} = e \right\}$$

$$\left\{ \because ae = a \right\}$$

Theorem-2 (Uniqueness of Identity): Prove that the identity element in a group is unique.

Proof

Suppose e and e' are two identity elements in a group G

$\forall a \in G$

We have

$$ae = a = ea \quad \text{--- (1)}$$

$$ae' = a = e'a \quad \text{--- (2)}$$

From (1) and (2)

$$ae = ae'$$

By left cancellation law

$$e = e'$$

Also

$$ea = e'a$$

By Right cancellation law

$$e = e'$$

Hence, identity element in a group is
unique

Theorem-3 (Uniqueness of inverse): Prove that the inverse of each element of a group is unique.

Proof

Let b and c are two inverse of same element a in a group G

$$ab = e = ba \quad \text{--- ①}$$

$$ac = e = ca \quad \text{--- ②}$$

From ① and ②

$$ab = ac$$

$$\boxed{b=c} \quad \left\{ \text{By Left cancellation law} \right\}$$

Also

$$ba = ca$$

$$\boxed{b=c} \quad \left\{ \text{By Right cancellation law} \right\}$$

Hence,

Inverse of each element is unique

VIMP
Theorem - 4 : Prove that $(a^{-1})^{-1} = a$

$\forall a \in G$

AKTU-2020-21

Proof

If $a \in G$, then $a^{-1} \in G$

We know that

$$a^{-1}a = e$$

$$(a^{-1})^{-1}(a^{-1}a) = (a^{-1})^{-1}e \quad \left\{ \begin{array}{l} \text{Pre-multiply by } (a^{-1})^{-1} \text{ on both} \\ \text{side} \end{array} \right\}$$

$$\{(a^{-1})^{-1}(a^{-1})\}a = (a^{-1})^{-1}e \quad \left\{ \begin{array}{l} \text{By Associative law} \end{array} \right\}$$

$$ea = (a^{-1})^{-1}e \quad \left\{ \begin{array}{l} a^{-1}a = e \end{array} \right\}$$

$$a = (a^{-1})^{-1} \quad \left\{ \because ae = ea = a \right\}$$

$$(a^{-1})^{-1} = a$$

Hence proved

Theorem-5 (Reversal Law) : Prove that $(ab)^{-1} = b^{-1}a^{-1}$
OR

$\forall a, b \in G \quad a^{-1} = b$
 $ab = e$

Prove that $(a * b)^{-1} = b^{-1} * a^{-1}$

$\forall a, b \in G$ (AKTU - 2020 - 21)

Proof

$\forall a, b \in G$

$aa^{-1} = e = a^{-1}a$

$$\begin{aligned} (ab)(b^{-1}a^{-1}) &= \{(ab)b^{-1}\}a^{-1} \quad \{\text{By Associative law}\} \\ &= \{a(bb^{-1})\}a^{-1} \quad \{\text{By Associative law}\} \\ &= (ae)a^{-1} \quad \left\{ \begin{array}{l} \because aa^{-1} = e \\ ae = e \end{array} \right\} \\ &= aa^{-1} \end{aligned}$$

$(ab)(b^{-1}a^{-1}) = e \quad \text{--- ①}$

Also

$$\begin{aligned} (b^{-1}a^{-1})(ab) &= b^{-1}\{a^{-1}(ab)\} \\ &\quad \{\text{By Associative law}\} \\ &= b^{-1}\{(a^{-1}a)b\} \\ &\quad \{\text{By associative law}\} \end{aligned}$$

$$(b^{-1}a^{-1})(ab) = b^{-1}(eb) \quad \left\{ \because a^{-1}a = e \right\}$$

$$= b^{-1}b \quad \left\{ \because eb = b \right\}$$

$$= e \quad \left\{ \because b^{-1}b = e \right\}$$

$$(b^{-1}a^{-1})(ab) = e \quad \text{--- (2)}$$

It is clear From (1) and (2)

$$(ab)^{-1} = b^{-1}a^{-1}$$

Hence proved

Proof

Let a and b be any two elements of G

Such that

$$a^{-1} = a$$

$$b^{-1} = b$$

Also

$$ab \in G$$

$$\Rightarrow (ab)^{-1} = ab$$

$$b^{-1}a^{-1} = ab \left\{ \because (ab)^{-1} = b^{-1}a^{-1} \right\}$$

$$ba = ab \text{ (Given)}$$

Hence group G satisfied commutative property

$\therefore G$ is an Abelian group

Theorem - 7 If an element a of a group G is of order n , then $a^m = e$ iff n is a divisor of m

Proof : Given $a^n = e$

Suppose n is a divisor of m . Then there exists an integer q such that

$$m = nq$$

$$a^m = a^{nq}$$

$$= (a^n)^q$$

$$= (e)^q \quad \{\text{Given}\}$$

$$= e$$

$$\boxed{a^m = e} \quad \text{Hence proved}$$

conversely suppose that $a^m = e$

$\because m$ and n are integers

$\therefore$ By Division algorithm, there exists two integers q and r such that

$$m = nq + r \quad \text{--- (1)}$$

$$a^m = e \quad (\text{Given})$$

$$a^{nq+r} = e$$

$$(a^{nq}) \times a^r = e$$

$$(a^n)^q \times a^x = e$$

$$(e)^q \times a^x = e$$

$$e \times a^x = e$$

$$a^x = e$$

But $0 \leq x < n$

It is only possible

When $x = 0$

From ①

$$m = nq$$

Hence

n is a divisor of m

Gateway Classes: 7455 9612 84

Theorem - 8 If a and b are arbitrary elements of a group G , then $(ab)^2 = a^2b^2$ iff G is an abelian group

Proof

Let G be a group and $a, b \in G$

Suppose $(ab)^2 = a^2b^2 \quad \forall a, b \in G$

$$(ab)^2 = a^2b^2$$

$$(ab)(ab) = (aa)(bb)$$

$$\{(ab)a\}b = \{(aa)b\}b \quad \{ \text{By Associative law} \}$$

$$\cancel{a}(ba)\cancel{b} = \cancel{a}(ab)\cancel{b} \quad \{ \text{By Associative law} \}$$

$$ba = ab \quad \left\{ \begin{array}{l} \text{By Left and Right} \\ \text{cancellation law} \end{array} \right\}$$

$$\boxed{ba = ab} \quad \forall a, b \in G \quad (\text{AKTU-2021-22})$$

$\Rightarrow G$ is an abelian group

conversely, suppose that G is an abelian then to show that

$$(ab)^2 = a^2b^2 \quad \forall a, b \in G$$

L.H.S

$$(ab)^2 = (ab)(ab)$$

$$= a\{b(ab)\} \quad \{ \text{By Associative law} \}$$

Note – (1) Any group of order less than 6 must be abelian.

(2) $\forall a \in G$, if $a^2 = e$ then G is an abelian group.

(3) If G is an abelian group then $(ab)^n = a^n b^n \forall a, b \in G$ and $\forall n \in \mathbb{I}$

(4) If a and x are arbitrary element of a group, then order of a is the same as the order of $x^{-1}ax$

i.e. $O(a) = O(x^{-1}ax)$

Q.1 Prove that $(ab)^{-1} = b^{-1}a^{-1} \quad \forall a, b \in G$

(AKTU – 2020 – 21)

Q.2 If a and b are arbitrary elements of a group G , then $(ab)^2 = a^2b^2$ iff G is an abelian group

(AKTU-2021-22)

Q3 – Find the order of each element of the group $G = \{0, 1, 2, 3, 4, 5\}$ under 'addition modulo 6'.

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

DISCRETE MATHEMATICS

UNIT -4 : Algebraic Structures

Lecture - 08

Today's Target

- *Sub – Group*
- *Cyclic Group*
- DPP
- PYQ

Gateway Classes : 7455 9612 84

$$H \subseteq G$$

SUBGROUP

$$(H, *)$$

$$(G, *)$$

Let H be a non – empty subset of a group G . Then H is called subgroup of group G if H is itself a group under the same operation as defined on G .

Note

- (i) The identity element of H is the same as that of G .
- (ii) The inverse of any element a of H is the same as the inverse of a in G .
- (iii) The order of any element a of H is same as the order of a in G .

Proper and improper subgroup

Every group G of order greater than one has atleast two subgroups which are

- (i) G (itself)
- (ii) $\{e\}$ i.e. the group of identity alone

The above two subgroups are known as improper or trivial subgroups. Any subgroup other than these two is known as a proper subgroup

Example :

(1) $(\mathbb{Z}, +)$ is a subgroup of $(\mathbb{Q}, +)$ and $(\mathbb{R}, +)$ $\mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$

(2) $(\mathbb{Q}, +)$ is a subgroup of $(\mathbb{R}, +)$

(3) $(\mathbb{E}, +)$ is a subgroup of $(\mathbb{Z}, +)$ and $(\mathbb{Q}, +)$

(4) $(\mathbb{O}, +)$ is a not subgroup of $(\mathbb{Z}, +)$ $\mathbb{O} \subseteq \mathbb{Z}$

(5) $H = \{2n : n \in \mathbb{Z}\}$ is a subgroup of $(\mathbb{Z}, +)$ $H \subseteq \mathbb{Z}$

(6) $H = \{3n : n \in \mathbb{Z}\}$ is a subgroup of $(\mathbb{Z}, +)$ $H \subseteq \mathbb{Z}$

(7) $(\mathbb{Q}_0, \times)$ is a subgroup of $(\mathbb{R}_0, \times)$

(8) $\{1, -1\}$ is a subgroup of $\{1, -1, i, -i\}$ under multiplication

Theorem - 1

The necessary and sufficient condition for a non - empty subset H of a group G to be a subgroup is $a \in H, b \in H \Rightarrow ab^{-1} \in H$

OR

A non - empty subset H of a group G is a subgroup of G iff $a \in H, b \in H \Rightarrow ab^{-1} \in H$ (AKTU - 2017)

Necessary Condition

Let H be a Sub-Group of G

and $a \in H, b \in H$

To Prove

$$a \in H, b \in H \Rightarrow ab^{-1} \in H$$

since $b \in H \Rightarrow b^{-1} \in H$ { By inverse property }

$$\therefore a \in H, b \in H \Rightarrow a, b^{-1} \in H$$

$$\Rightarrow ab^{-1} \in H \text{ { By closure property } }$$

Hence

$$a \in H, b \in H \Rightarrow ab^{-1} \in H$$

Sufficient condition

Let H be non-empty subset of G
Such that

$$a \in H, b \in H \Rightarrow ab^{-1} \in H \quad \text{--- ①}$$

To Prove

H is a Sub-group of G

(i) Identity Property

From ①

$$b \in H, b \in H \Rightarrow bb^{-1} \in H$$

$$\Rightarrow e \in H$$

$\Rightarrow H$ contain identity element

$\therefore$ Satisfied identity property

(ii) Inverse property

$$\therefore e \in H \text{ (By identity property)}$$

From ①

$$e \in H, b \in H \Rightarrow eb^{-1} \in H$$

$$\Rightarrow b^{-1} \in H \quad \{ae = a\}$$

$\Rightarrow H$ contain inverse of each element

$\therefore$ Satisfied inverse property

(iii) Closure Property

$\therefore b^{-1} \in H$ { By inverse property }

From ①

$$a \in H, b^{-1} \in H \Rightarrow a(b^{-1})^{-1} \in H$$
$$\Rightarrow ab \in H$$

$\Rightarrow H$ is closed under multiplication

$\therefore$ Satisfied closure property

$$\{(a^{-1})^{-1} = a\}$$

Associative Property

Let $a, b, c \in H$

$$\Rightarrow a, b, c \in G \{ \because H \subseteq G \}$$

But group satisfied Associative property

$$\Rightarrow (a \cdot b) \cdot c = a \cdot (b \cdot c) \quad \forall a, b, c \in H$$

$\therefore$ Satisfied associative property

Hence

H is a Subgroup of G

Remember

A non – empty subset H of a group G is a subgroup of G iff

(i) $a \in H, b \in H \Rightarrow ab^{-1} \in H$ (For multiplication)

(ii) $a \in H, b \in H \Rightarrow a - b \in H$ (For addition)

Q.1 Prove that $\{1, -1\}$ is a subgroup of group $\{1, -1, i, -i\}$ under multiplication

Let $H = \{1, -1\}$ and $G = \{1, -1, i, -i\}$

$$\Rightarrow H \subseteq G$$

Let $a \in H, b \in H$ such that

$$a = 1, b = -1$$

$$b^{-1} = -1$$

$$ab^{-1} = 1 \times (-1) = -1$$

$$\Rightarrow ab^{-1} \in H$$

$$\Rightarrow a \in H, b \in H \Rightarrow ab^{-1} \in H$$

Hence

H is a subgroup of G

Q.2 Prove that $H = \{3n : n \in \mathbb{Z}\}$ is a subgroup of integers under addition

$$H = \{3n : n \in \mathbb{Z}\}$$

$$H = \{- \dots -6, -3, 0, 3, 6 \dots\}$$

$$\mathbb{Z} = \{- \dots -3, -2, -1, 0, 1, 2, 3, 4, 5, 6 \dots\}$$

$$\Rightarrow H \subseteq \mathbb{Z}$$

Let $a \in H, b \in H$ such that

$$a = -6, b = 3$$

$$a - b = -6 - 3 = -9 \in H$$

$$\therefore a \in H, b \in H \Rightarrow (a - b) \in H$$

Hence

H is a subgroup of integers

Prove that the intersection of two subgroups of a group G , is also subgroup of G .

OR

If H_1 and H_2 are two subgroups of a group G then $H_1 \cap H_2$ is also a subgroup of G . (AKTU – 2019)

Let H_1 and H_2 are two sub-groups $\Rightarrow H_1 \cap H_2 \neq \emptyset$
of a group G

To Prove

$H_1 \cap H_2$ is a sub-group of G

$\because e \in H_1$ and $e \in H_2$

$\Rightarrow e \in H_1 \cap H_2$

Let $a, b \in H_1 \cap H_2$

$\Rightarrow a, b \in H_1$ and $a, b \in H_2$

$\Rightarrow ab^{-1} \in H_1$ and $ab^{-1} \in H_2$

$\left\{ \because H_1 \text{ and } H_2 \text{ are sub-groups} \right\}$

$$\Rightarrow ab^{-1} \in H_1 \cap H_2$$

$$\therefore a \in H_1 \cap H_2, b \in H_1 \cap H_2 \Rightarrow ab^{-1} \in H_1 \cap H_2$$

Hence

$H_1 \cap H_2$ is a sub-group of group G

Generalisation

If $H_1, H_2, H_3, \dots, H_n$ are finite sub-groups of G then

$H_1 \cap H_2 \cap H_3 \cap \dots \cap H_n$ is also a sub-group of G

(1) Union of two subgroups is not necessarily a subgroup.

We know that

The group $G = (\mathbb{Z}, +)$ has two sub-groups

$$H_1 = \{2n : n \in \mathbb{Z}\}$$

$$H_2 = \{3n : n \in \mathbb{Z}\}$$

$$H_1 \cup H_2 = \{\dots, -6, -4, -3, -2, 0, 2, 3, 4, 6, \dots\}$$

Since

$$2 \in H_1 \cup H_2 \text{ and } 3 \in H_1 \cup H_2$$

But

$$2 + 3 = 5 \notin H_1 \cup H_2$$

$\Rightarrow H_1 \cup H_2$ is not closed

under addition

Hence

$H_1 \cup H_2$ is not a sub-group

(2) The union of two subgroups of a group G is a subgroup of G iff one is contained in the other.

CYCLIC GROUP

- Let G be any group. If for $a \in G$, every element $x \in G$ can be generated by a such that

$$x = a^n, n \in I.$$

Then G is called a cyclic group and a is called its generator

- A cyclic group G with generator a is denoted by

$$G = \langle a \rangle \text{ OR } G = \{a\}$$

Note

- If a is a generator then its inverse a^{-1} is also a generator.
- if $\text{order of an element } a = \text{order of group}$ then a is called generator

Gateway Classes : 7455 9612 84

Q.1 Prove that the multiplicative group of the cube roots of unity $\{1, \omega, \omega^2\}$ is a cyclic group.

Find its generator.

$$\text{Let } G = \{1, \omega, \omega^2\}$$

$$G = \{\omega^3, \omega, \omega^2\}$$

$$\Rightarrow G = \langle \omega \rangle$$

$\Rightarrow G$ is a cyclic group

Also

$$G = \{(\omega^2)^3, (\omega^2)^2, \omega^2\}$$

$$G = \langle \omega^2 \rangle$$

$\Rightarrow G$ is cyclic group with two generators ω and ω^2

Q.2 Prove that $G = \{1, -1, i, -i\}$ is cyclic. Also find its generator.

$$G = \{1, -1, i, -i\}$$

$$G = \{i^4, i^2, i, i^3\}$$

$$\Rightarrow G = \langle i \rangle$$

Also

$$G = \{(-i)^4, (-i)^2, (-i)^3, -i\}$$

$$\Rightarrow G = \langle -i \rangle$$

Hence, G is a cyclic group with generators i and $-i$

Gateway Classes: 7455 9642 84

Q.3 Show that the group $G = (\{1, 2, 3, 4, 5, 6\}, X_7)$ is cyclic. How many generators are there.

$$G = \{1, 2, 3, 4, 5, 6\}$$

$$3^1 = 3$$

$$3^2 = 3 \times_7 3 = 2$$

$$3^3 = 3^2 \times_7 3 = 2 \times_7 3 = 6$$

$$3^4 = 3^3 \times_7 3 = 6 \times_7 3 = 4$$

$$3^5 = 3^4 \times_7 3 = 4 \times_7 3 = 5$$

$$3^6 = 3^5 \times_7 3 = 5 \times_7 3 = 1$$

$$\Rightarrow G = \langle 3 \rangle$$

Since

$$3^{-1} = 5$$

$$\Rightarrow G = \langle 5 \rangle$$

Hence

G is a cyclic group with generators
3 and 5

Theorem - 1

Prove that every cyclic group is an abelian group.

(AKTU - 2020)

Let $G = \langle a \rangle$ be a cyclic group and

$$x, y \in G$$

To Prove

G is an abelian group

$\because a$ is the generator of the group

$$\therefore x = a^m, y = a^n \quad \text{Where } m, n \in \mathbb{Z}$$

$$\Rightarrow xy = a^m \times a^n$$

$$\Rightarrow xy = a^{m+n}$$

$$\Rightarrow xy = a^{n+m} \quad \left\{ \because m+n = n+m \right\}$$

$$\Rightarrow xy = a^n \times a^m$$

$$\Rightarrow xy = yx \quad \forall x, y \in G$$

$\Rightarrow$ group G satisfied commutative property

Hence, G is an Abelian group

Theorem - 2

If a is a generator of a cyclic group G , then a^{-1} is also a generator of G .

Let $G = \langle a \rangle$ be a cyclic group with generator a and $x \in G$

So, there exists an integer n such that

$$x = a^n = a^{(-1)(-n)}$$

$$x = (a^{-1})^{-n}$$

$\Rightarrow$ Every Element of G is generated by a^{-1}

Hence a^{-1} is also a generator of group G

Theorem – 3

Every sub group of a cyclic group is cyclic

Theorem – 4

Every group of prime order is cyclic

Gateway Classes :: 7455 9612 84

Topic : Sub – Group and Cyclic Group

Q.1 The necessary and sufficient condition for a non – empty subset H of a group G to be a subgroup is $a \in H, b \in H \Rightarrow ab^{-1} \in H$ (AKTU – 2017)

OR

A non – empty subset H of a group G is a subgroup of G iff $a \in H, b \in H \Rightarrow ab^{-1} \in H$

Q.2 Prove that the intersection of two subgroups of a group G , is also subgroup of G .

OR

If H_1 and H_2 are two subgroups of a group G then $H_1 \cap H_2$ is also a subgroup of G . (AKTU-2019)

Q.3 Prove that every cyclic group is an abelian group. (AKTU – 2020)

Q.4 Prove that $(G = \{0, 1, 2, 3, 4, 5\}, +_6)$ is cyclic group. Find generator .

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

DISCRETE MATHEMATICS

UNIT -4 : Algebraic Structures

Lecture - 09

Today's Target

- *COSET*
- *Lagrange's Theorem*
- DPP
- PYQ

Gateway Classes : 7455 9612 84

COSETS (For Multiplication)

Let H be a subgroup of group G and let $a \in G$ then the set.

$aH = \{ah : h \in H\}$ is called left cosets of H in G

$Ha = \{ha : h \in H\}$ is called Right coset of H in G

COSETS (For Addition)

Let H be a subgroup of G and ~~let $a \in G$ and~~ let $a \in G$ then the set

$a + H = \{a + h : h \in H\}$ is called left coset of H in G

$H + a = \{h + a : h \in H\}$ is called Right coset of H in G

Gateway Classes : 7455 9612 84

Properties of cosets

✓ (1) If $a \in G$ as well as $a \in H$, then

$$aH = Ha = H$$

$$H \subseteq G$$

✓ (2) If e be the identity element of G , then

$$He = eH = H$$

Hence H itself is a left as well as right coset of H in G

✓ (3) Any two left or Right cosets of H is either disjoint or identical

✓ (4) The union of all left cosets or right cosets is equal to G

Q.1 Let $G = \{1, -1, i, -i\}$ be a multiplicative group and $H = \{1, -1\}$ is a subgroup of G .

Find all distincts left and right cosets of H in G .

$$H = \{1, -1\} \text{ and } G = \{1, -1, i, -i\}$$

$$1H = \{1, -1\} = H \quad H1 = \{1, -1\} = H$$

$$-1H = \{-1, 1\} = H \quad H(-1) = \{-1, 1\} = H$$

$$iH = \{i, -i\}$$

$$Hi = \{i, -i\}$$

$$-iH = \{-i, i\}$$

$$H(-i) = \{-i, i\}$$

Hence

Two Distincts left / Right cosets

Union of Left cosets

$$1H \cup (-1)H \cup iH \cup (-i)H = \{1, -1, i, -i\}$$

$$G = \{1, -1, i, -i\}$$

Union of Right cosets

$$H1 \cup H(-1) \cup Hi \cup H(-i) = \{1, -1, i, -i\}$$

$$G = \{1, -1, i, -i\}$$

Q.2 What do you mean by cosets of a subgroup? Consider the group Z of integers under addition and the subgroup $H = \{\dots, -12, -6, 0, 6, 12, \dots\}$ consisting of multiple of 6.

(i) Find the cosets of H in Z .

(ii) What is index of H in Z ?

(AKTU 2019-20, Marks 10)

$$Z = \{\dots, -6, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6, \dots\}$$

$$H = \{\dots, -12, -6, 0, 6, 12, \dots\}$$

(i) Left cosets of H

$$0 + H = \{\dots, -12, -6, 0, 6, 12, \dots\}$$

$$1 + H = \{\dots, -11, -5, 1, 7, 13, \dots\}$$

$$2+H = \{ \dots -10, -4, 2, 8, 14 \dots \}$$

$$3+H = \{ \dots -9, -3, 3, 9, 15 \dots \}$$

$$4+H = \{ \dots -8, -2, 4, 10, 16 \dots \}$$

$$5+H = \{ \dots -7, -1, 5, 11, 17 \dots \}$$

$$6+H = \{ \dots -6, 0, 6, 12, 18 \dots \}$$

Now, Repetition Starts

Hence

$0+H, 1+H, 2+H, 3+H, 4+H, 5+H$ are 6

distinct coset of H in G

Also

Right cosets and Left cosets are similar

(ii) Index Represent the total number of distinct cosets

$$\text{Index of } H \text{ in } Z = 6$$

Lagrange's Theorem

Statement:

The order of each subgroup of a finite group is a divisor of the order of the group.

Proof:

Let G be a finite group of order n

$$O(G) = n$$

Where $G = \{a_1, a_2, a_3, \dots, a_n\}$

Let H be a sub-group of G of order m

$$O(H) = m$$

Where $H = \{h_1, h_2, h_3, \dots, h_m\}$

To show that

$O(H)$ is a Divisor of $O(G)$

Since $a_1 \in G$

$Ha_1 = \{h_1a_1, h_2a_1, h_3a_1, \dots, h_ma_1\}$

Ha_1 be a Right coset of H in G and

$h_1a_1, h_2a_1, h_3a_1, \dots, h_ma_1$ are

m distinct elements

$$\therefore O(Ha_1) = m$$

Similarly

$$O(Ha_2) = m$$

$$O(Ha_3) = m$$

⋮

$$O(Ha_n) = m$$

Suppose there are k distinct cosets of H in G

As given below

$$Ha_1 = \{h_1a_1, h_2a_1, h_3a_1, \dots, h_ma_1\}$$

$$Ha_2 = \{h_1a_2, h_2a_2, h_3a_2, \dots, h_ma_2\}$$

$$Ha_3 = \{h_1a_3, h_2a_3, h_3a_3, \dots, h_ma_3\}$$

⋮

$$Ha_k = \{h_1a_k, h_2a_k, h_3a_k, \dots, h_ma_k\}$$

We know that

The union of all Left or Right cosets of H is equal to the Group

$$G = Ha_1 \cup Ha_2 \cup Ha_3 \cup \dots \cup Ha_k$$

$$O(G) = O(Ha_1) + O(Ha_2) + O(Ha_3) + \dots + O(Ha_k)$$

$$n = m + m + m + \dots + m \text{ (k times)}$$

$$n = km$$

$$\frac{n}{m} = k$$

$$\frac{O(G)}{O(H)} = k$$

Where $k \in \mathbb{N}$

$\Rightarrow O(H)$ is a Divisor of $O(G)$

Hence proved

Topic : COSET and Lagrange's Theorem

Q.1 What do you mean by cosets of subgroup ? Consider the group Z of integers under addition and the subgroup $H = \{ \dots, -10, -5, 0, 5, 10, \dots \}$ considering the multiple of 5.

(i) Find the cosets of H in Z .

(ii) What is index of H in Z ?

Q.2 State and prove Lagrange theorem for group. (AKTU 2020-21, Marks 10)

Q.3 Explain cyclic group. Let H be a subgroup of a finite group G . Justify the statement "the order of H is a divisor of the order of G "

(AKTU 2021-22, Marks 10)

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

DISCRETE MATHEMATICS

UNIT -4 : Algebraic Structures

Lecture - 10

Today's Target

- *Normal Subgroup*
- DPP
- PYQ

Gateway Classes : 7455 9612 84

COSETS (For Multiplication)

Let H be a subgroup of group G and let $a \in G$ then the set.

$aH = \{ah : h \in H\}$ is called left cosets of H in G

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COSETS (For Addition)

Let H be a subgroup of G and let $a \in G$ and let $a \in G$ then the set

$a + H = \{a + h : h \in H\}$ is called left coset of H in G

$H + a = \{h + a : h \in H\}$ is called Right coset of H in G

Note : Left coset and Right coset of H may or may not be equal

Definition – 1

A Subgroup H of a group G is said to be a Normal Subgroup of G iff

$$Hx = xH \quad \forall x \in G$$

Definition – 2

A Subgroup H of a group G is said to be a Normal Subgroup of G iff

$$x h x^{-1} \in H \quad \forall h \in H \text{ and } x \in G$$

From this definition it is clear that H is a normal Subgroup of G iff

$$x H x^{-1} \subseteq H \quad \forall x \in G$$

Definition – 3

A Subgroup H of a group G is said to be a normal Subgroup of G iff

$$x H x^{-1} = H \quad \forall x \in G$$

Theorem – 1

Prove that every subgroup of an abelian group is normal subgroup.

Let H be a subgroup of any abelian group G .

If $\forall h \in H$ and $x \in G$, then

$$xhx^{-1} = (xh)x^{-1}$$

$$= (hx)x^{-1} \quad \{\text{By commutative property}\}$$

$$= h(xx^{-1}) \quad \{\text{By associative property}\}$$

$$= he \quad \{\text{By inverse property}\}$$

$$= h \quad \{\text{By identity property}\}$$

$$= h \in H$$

$$\Rightarrow xhx^{-1} \in H$$

$\therefore H$ is a normal subgroup G

Note

Every subgroup of a cyclic group is normal subgroup.

Theorem – 2

Prove that the intersection of two normal subgroup of a group is a normal subgroup. (AKTU-2020)

Let H and K be two normal subgroup of a group G

To Prove

$H \cap K$ is also a normal subgroup

We know that

The intersection of two subgroup is also a Subgroup

$\therefore H \cap K$ is a subgroup of G

Let $x \in G$ and $h \in H \cap K$

$\therefore h \in H \cap K$

$\therefore h \in H, h \in K$

$\therefore H$ is a normal subgroup

$$xhx^{-1} \in H \quad \forall x \in G \text{ and } h \in H$$

$\therefore K$ is a also a normal subgroup

$$xhx^{-1} \in K \quad \forall x \in G \text{ and } h \in K$$

$$\Rightarrow xhx^{-1} \in H \text{ and } xhx^{-1} \in K$$

$$\Rightarrow xhx^{-1} \in H \cap K \quad \forall x \in G \text{ and } h \in H \cap K$$

Hence

$H \cap K$ is a normal subgroup

Theorem – 3

A subgroup H of a group G is normal subgroup if and only if $x^{-1}h x \in H \quad \forall h \in H$ and $x \in G$ (AKTU-2021)

Let H be a normal subgroup

Prove that

$$x^{-1}h x \in H \quad \forall h \in H \text{ and } x \in G$$

We know that for a normal subgroup

$$H x = x H$$

$$\Rightarrow h x = x h$$

$$\Rightarrow h x x^{-1} = x h x^{-1}$$

$$\Rightarrow h e = x h x^{-1} \quad \{\text{By inverse property}\}$$

$$\Rightarrow h = x h x^{-1} \quad \{\text{By identity property}\}$$

$$\Rightarrow x h x^{-1} \in H$$

Conversely,

$$\text{Let } x h x^{-1} \in H$$

$$\forall x \in G \text{ and } h \in H$$

$$x h x^{-1} \in H$$

$$\Rightarrow x H x^{-1} \subseteq H$$

Hence,

H is a normal subgroup.

Theorem — 4

A subgroup H of a group G is normal iff $x H x^{-1} = H \quad \forall x \in G$

Let

$$x H x^{-1} = H \quad \forall x \in G$$

$$\Rightarrow x H x^{-1} \subseteq H$$

$\therefore H$ is a normal subgroup of G

Conversely, Let H be a normal subgroup of G then $\forall x \in G$

$$x H x^{-1} \subseteq H \quad \text{--- (1)}$$

$$\because x \in G \Rightarrow x^{-1} \in G$$

Using (1)

$$x^{-1} H (x^{-1})^{-1} \subseteq H$$

$$x^{-1} H x \subseteq H \quad \{\because (x^{-1})^{-1} = x\}$$

$$x x^{-1} H x \subseteq x H$$

$$x x^{-1} H x x^{-1} \subseteq x H x^{-1}$$

$$e H e \subseteq x H x^{-1} \quad \{\text{By inverse property}\}$$

$$H e \subseteq x H x^{-1} \quad \{\text{By identity property}\}$$

$$H \subseteq x H x^{-1} \quad \text{--- (2)}$$

Using (1) and (2)

$$x H x^{-1} = H \quad \forall x \in G$$

Hence proved

Theorem – 5

A subgroup H of a group G is normal subgroup iff Left coset of H in G is a right coset of H in G

Let H is a normal subgroup of G

To show that $xH = Hx \quad \forall x \in G$

$$\Rightarrow xHx^{-1} = He$$

$$\Rightarrow xHx^{-1} = H \quad \forall x \in G$$

Since H is normal subgroup

$\therefore H$ is a normal subgroup

$$xHx^{-1} = H \quad \forall x \in G$$

$$\Rightarrow xHx^{-1}x = Hx$$

$$\Rightarrow xHe = Hx$$

$$\Rightarrow xH = Hx$$

Conversely, Let $xH = Hx \quad \forall x \in G$

$$\Rightarrow (xH)x^{-1} = Hxx^{-1}$$

Q. 1 What is normal Subgroup?

Q.2 Prove that the intersection of two normal subgroup of a group is a normal subgroup. (AKTU-2020)

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

DISCRETE MATHEMATICS

UNIT -4 : Algebraic Structures

Lecture - 11

Today's Target

- *Permutation Group / Symmetric Group*
- DPP
- PYQ

Permutation

Let $S = \{a_1, a_2, a_3, \dots, a_n\}$ be a finite set of n elements.

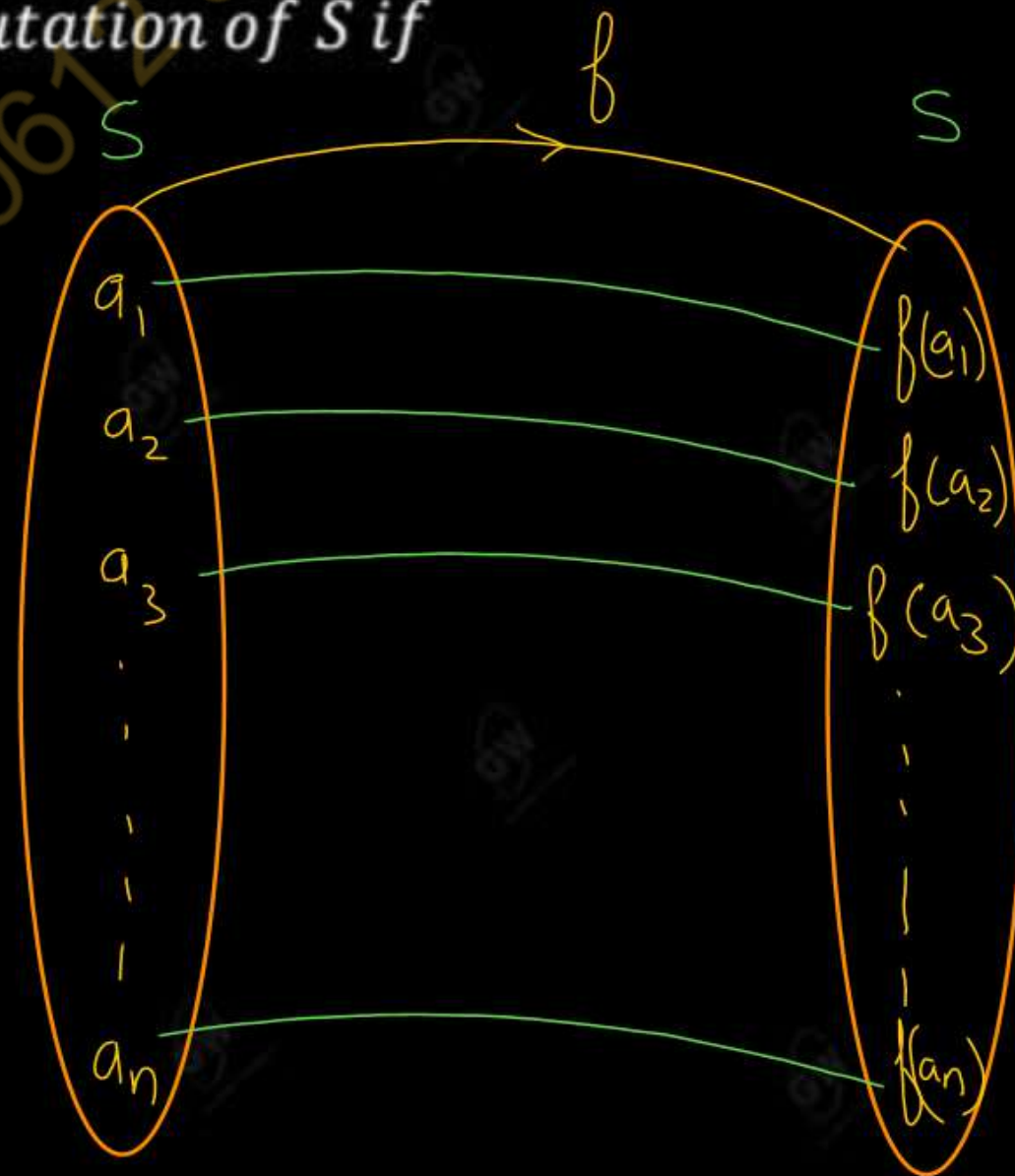
Then a function/Mapping $f: S \rightarrow S$ is said to be a permutation of S if

(i) f is one – one

(ii) f is onto

Notation of permutation f

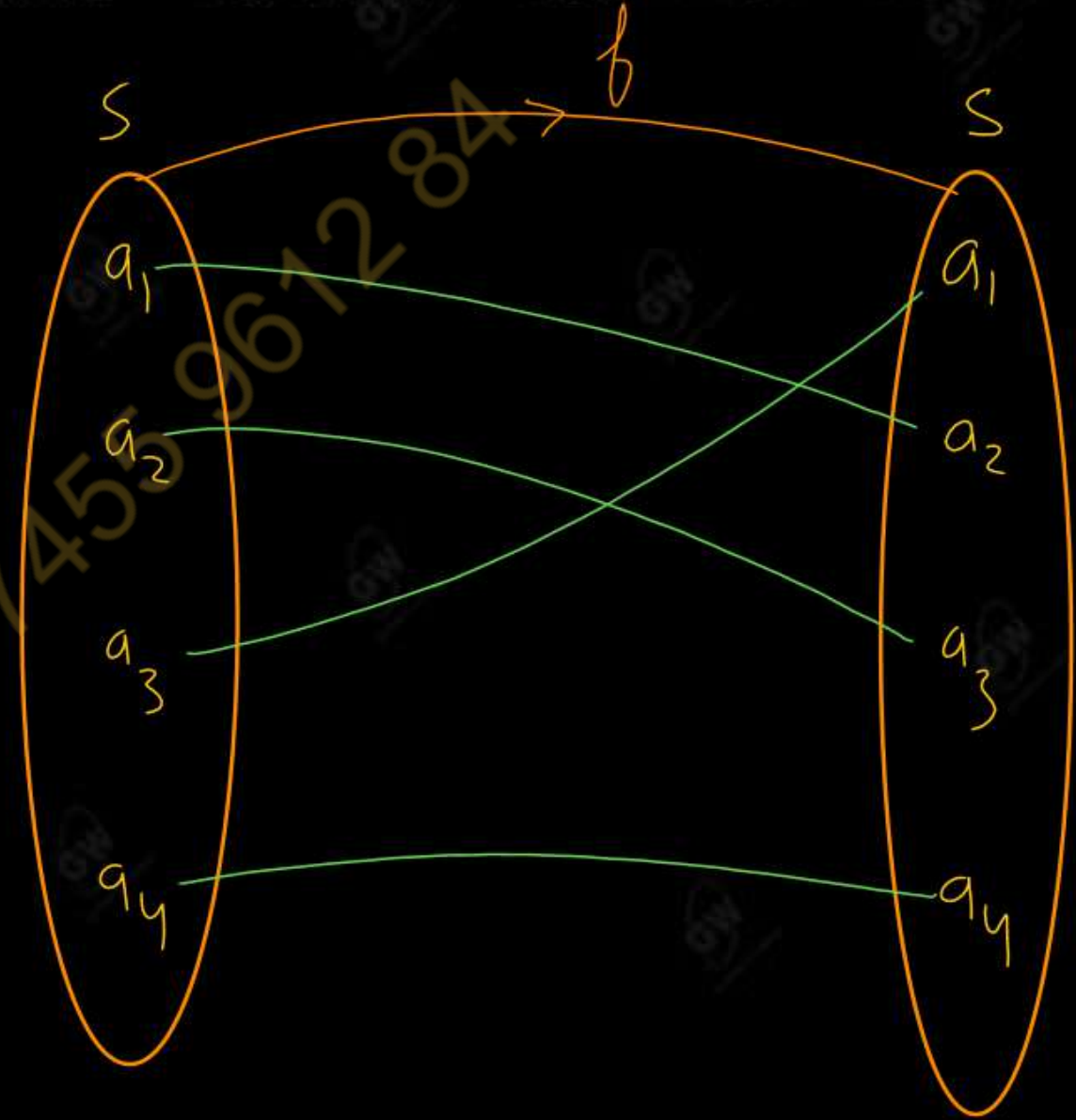
$$f = \begin{pmatrix} a_1 & a_2 & a_3 & \dots & a_n \\ f(a_1) & f(a_2) & f(a_3) & \dots & f(a_n) \end{pmatrix}$$



Q.1 Let $S = \{a_1, a_2, a_3, a_4\}$ and $f(a_1) = a_2, f(a_2) = a_3, f(a_3) = a_1, f(a_4) = a_4$ find permutation f

$$f = \begin{pmatrix} a_1 & a_2 & a_3 & a_4 \\ f(a_1) & f(a_2) & f(a_3) & f(a_4) \end{pmatrix}$$

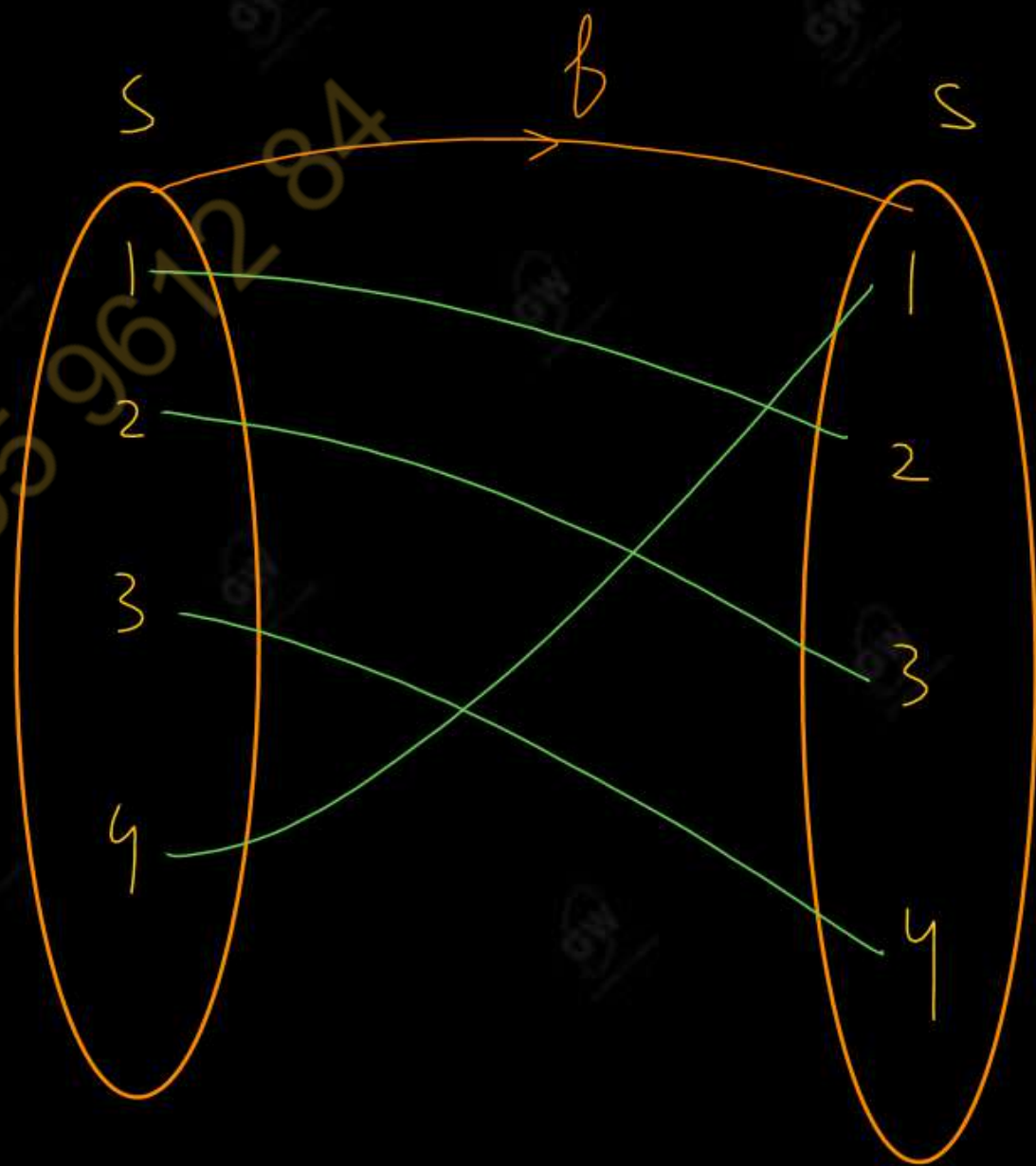
$$f = \begin{pmatrix} a_1 & a_2 & a_3 & a_4 \\ a_2 & a_3 & a_1 & a_4 \end{pmatrix}$$



Q.2 Let $S = \{1, 2, 3, 4\}$ and $f(1) = 2, f(2) = 3, f(3) = 4, f(4) = 1$ find permutation

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ f(1) & f(2) & f(3) & f(4) \end{pmatrix}$$

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$$



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Note: (1) The order of the column in permutation is immaterial.

For example If $f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$ and $g = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix}$

Then f and g are same.

(2) If $S = \{a_1, a_2, a_3, \dots, a_n\}$ be a finite set with n elements then

Total number of permutation on set $S = n!$

Equality of two Permutation

Let f and g be two permutations on non – empty Set S .

Then $f = g$ if and only if

$$f(a) = g(a) \quad \forall x \in S$$

Q.3 Check whether the permutation f and g are equal or not

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \text{ and } g = \begin{pmatrix} 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix}$$

$$f(1) = 2 = g(1)$$

Hence f and g are equal

$$f(2) = 3 = g(2)$$

$$f(3) = 4 = g(3)$$

$$f(4) = 1 = g(4)$$

$$\Rightarrow f(a) = g(a) \quad \forall a \in S$$

Composition of Permutation (Product of permutation)

The Composition of two permutations f and g of the same degree is denoted by $f \circ g$ or fg .

$f \circ g$ is also a permutation.

Note: If $g \circ f$ is defined then $g \circ f$ is also a permutation.

Q.4 Find the product of two permutations and show that it is not commutative

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \text{ and } g = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 4 \end{pmatrix}$$

$$f \circ g = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 4 \end{pmatrix}$$

$$f \circ g = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$$

$$g \circ f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 4 \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix}$$

$$g \circ f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix}$$

$$\Rightarrow f \circ g \neq g \circ f$$

Not commutative

Identity Permutation

Let $S = \{a_1, a_2, a_3, \dots, a_n\}$ then

$f = \begin{pmatrix} a_1 & a_2 & a_3 & \dots & a_n \\ a_1 & a_2 & a_3 & \dots & a_n \end{pmatrix}$ is called identity permutation on S and denoted by I

Q.5 How many times $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix}$ be multiplied to itself to produce identity permutation I ?

$$\text{Let } f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix}$$

$$f^2 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix}$$

$$f^2 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix}$$

$$f^3 = f^2 f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix}$$

$$f^3 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{pmatrix}$$

3 times

Inverse Permutation

Since a permutation is one one on to map and hence it is invertible

If $f = \begin{pmatrix} a_1 & a_2 & a_3 & \dots & a_n \\ b_1 & b_2 & b_3 & \dots & b_n \end{pmatrix}$ then

its unique inverse permutation is denoted by f^{-1} and given as

$$f^{-1} = \begin{pmatrix} b_1 & b_2 & b_3 & \dots & b_n \\ a_1 & a_2 & a_3 & \dots & a_n \end{pmatrix}$$

Such that

$$f \circ f^{-1} = f f^{-1} = I$$

$$f^{-1} \circ f = f^{-1} f = I$$

Gateway Classes : 7455 9612 84

Q.6 Find the inverse permutation of $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 5 & 4 \end{pmatrix}$

$$\text{Let } f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 5 & 4 \end{pmatrix}$$

$$f^{-1} = \begin{pmatrix} 2 & 3 & 1 & 5 & 4 \\ 1 & 2 & 3 & 4 & 5 \end{pmatrix}$$

$$f^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 1 & 2 & 5 & 4 \end{pmatrix}$$

Permutation Group / Symmetric Group

Let S be a Set of n elements. The set of all one – one onto mapping from S to S form a group with respect to the composition of mapping.

*The set of permutation (mapping) is called permutation group or Symmetric Group of degree n and denoted by S_n or $(S_n, *)$*

Order of permutation group

$$O(S_n) = n !$$

Gateway Classes : 7455 9672 84

Q.7 Find the elements and multiplication table of symmetric group S_3

$$S_3 = \{1, 2, 3\} \text{ and } O(S_3) = 3! = 6$$

$$S_3 = \{b_1, b_2, b_3, b_4, b_5, b_6\}$$

$$b_1 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}$$

$$b_2 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$$

$$b_3 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$$

$$b_4 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

$$b_5 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

$$b_6 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$$

Composition table

$\circ$	b_1	b_2	b_3	b_4	b_5	b_6
b_1	b_1	b_2	b_3	b_4	b_5	b_6
b_2	b_2	b_1	b_4	b_3	b_6	b_5
b_3	b_3	b_5	b_1	b_6	b_2	b_4
b_4	b_4	b_6	b_2	b_5	b_1	b_3
b_5	b_5	b_3	b_6	b_1	b_4	b_2
b_6	b_6	b_4	b_5	b_2	b_3	b_1

Theorem :

Prove that the set S_n of all permutation is a finite group of order $n!$ with respect to binary composition of permutation

Proof:

(i) Closure properties :

$$\forall f \in S_n \text{ \& } g \in S_n \\ \Rightarrow f \cdot g = f \circ g \in S_n$$

(ii) Associativity :

$$\forall f, g, h \in S_n$$

We have

$$(f \circ g) \circ h = f \circ (g \circ h)$$

So $(S_n, \circ)$ is a group under the composition operation and it is also called permutation group or symmetric group.

(iii) Identity :

$\forall f \in S_n$ there exists Identity

permutation $I_n \in S_n$ such that $f \circ I_n = f$

$I_n =$ identity element.

(iv) Inverse :

$\forall f \in S_n$ there exists inverse

permutation $f^{-1} \in S_n$ such that

$$f \circ f^{-1} = I_n$$

Cyclic Permutation

Let $S = \{a_1, a_2, a_3 \dots a_n\}$ be a finite Set of n elements, then the permutation

$$f = \begin{pmatrix} a_1 & a_2 & a_3 & \dots & a_n \\ a_2 & a_3 & a_4 & \dots & a_1 \end{pmatrix} = (a_1 \ a_2 \ a_3 \ \dots \ a_n) \text{ is called cyclic permutation}$$

Note

(i) In cyclic permutation

$$f(a_1) = a_2, \quad f(a_2) = a_3, \quad f(a_3) = a_4, \dots, f(a_n) = a_1$$

(ii) Self image element does not contain in cycle

if $f(a) = a$ then, $a \notin \text{cyclic}$

Length of cycle (n)

Number of elements in a cycle is called length of cycle

Order of cycle

$$O(f) = \text{Length of cycle}(n)$$

Q.8 Find cyclic permutation for the following permutation

$$(i) \quad f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$$

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$$

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 \end{pmatrix}$$

$$(ii) \quad f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 4 & 3 & 5 & 6 & 2 \end{pmatrix}$$

$$f = \begin{pmatrix} \overset{x}{1} & 2 & \overset{x}{3} & 4 & 5 & 6 \\ 1 & 4 & 3 & 5 & 6 & 2 \end{pmatrix}$$

$$f = \begin{pmatrix} 2 & 4 & 5 & 6 \\ 4 & 5 & 6 & 2 \end{pmatrix}$$

$$f = (2 \ 4 \ 5 \ 6)$$

Q.9 Let $A = \{1, 2, 3, 4, 5\}$ find $(1\ 3) \circ (2\ 4\ 5) \circ (2\ 3)$

$$(1\ 3) \circ (2\ 4\ 5) \circ (2\ 3)$$

$$= \begin{pmatrix} 1 & 3 & 2 & 4 & 5 \\ 3 & 1 & 2 & 4 & 5 \end{pmatrix} \circ \begin{pmatrix} 2 & 4 & 5 & 1 & 3 \\ 4 & 5 & 2 & 1 & 3 \end{pmatrix} \circ \begin{pmatrix} 2 & 3 & 1 & 4 & 5 \\ 3 & 2 & 1 & 4 & 5 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 3 & 2 & 4 & 5 \\ 3 & 1 & 4 & 5 & 2 \end{pmatrix} \circ \begin{pmatrix} 2 & 3 & 1 & 4 & 5 \\ 3 & 2 & 1 & 4 & 5 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 3 & 2 & 4 & 5 \\ 2 & 1 & 4 & 5 & 3 \end{pmatrix}$$

$$= (1\ 2\ 4\ 5\ 3)$$

- A cycle of length 2 of the form $(a\ b)$ is called transposition
- Every permutation can be express as the product of cyclic permutations
- Every cyclic permutation can be expressed as the product of transpositions
- Consider a cycle $(a_1, a_2, a_3 \dots a_n)$

$$(a_1\ a_2\ a_3 \dots a_n) = (a_1\ a_2)(a_1\ a_3)(a_1\ a_4) \dots (a_1\ a_n)$$

- **Number of transpositions = $n - 1$**

Where n is the length of cycle

Q.10 – Express the following permutation as the product of transposition

$$(i) \quad f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 3 & 1 & 4 & 6 & 7 & 8 & 5 \end{pmatrix}$$

$$f = (1 \ 2 \ 3)(5 \ 6 \ 7 \ 8)$$

$$f = (1 \ 2)(1 \ 3)(5 \ 6)(5 \ 7)(5 \ 8)$$

$$(ii) \quad f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 6 & 2 & 5 & 1 & 3 & 8 & 7 \end{pmatrix}$$

$$f = (1 \ 4 \ 5)(2 \ 6 \ 3)(7 \ 8)$$

$$f = (1 \ 4)(1 \ 5)(2 \ 6)(2 \ 3)(7 \ 8)$$

Gateway Classes

Even and odd Permutation

Even Permutation

A Permutation is said to be even permutation if it can be expressed as the product of even number of transposition

Odd Permutation

A Permutation is said to be odd permutation if it can be expressed as the product of odd number of transposition

Gateway Classes

7455 9612 84

HW

Q.11 Show that the permutation $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 5 & 6 & 2 & 4 & 1 & 3 \end{pmatrix}$ is odd, while the permutation $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 3 & 4 & 5 & 2 & 1 \end{pmatrix}$ is even

Gateway Classes :: 7455 9612 84

Topic : Permutation Group / Symmetric Group

Q. 1 Find the product of two permutations and show that it is not commutative

$$f = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \text{ and } g = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$$

Q. 2 Find cyclic permutation of the permutation $f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix}$

Q. 3 If $S = \{1, 2, 3, 4, 5, 6\}$ compute $(5, 6, 3) \circ (4, 1, 3, 5)$

Q. 4 Show that $f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 7 & 3 & 1 & 8 & 5 & 6 & 2 & 4 \end{pmatrix}$ is even

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

DISCRETE MATHEMATICS

UNIT -4 : Algebraic Structures

Lecture - 12

Today's Target

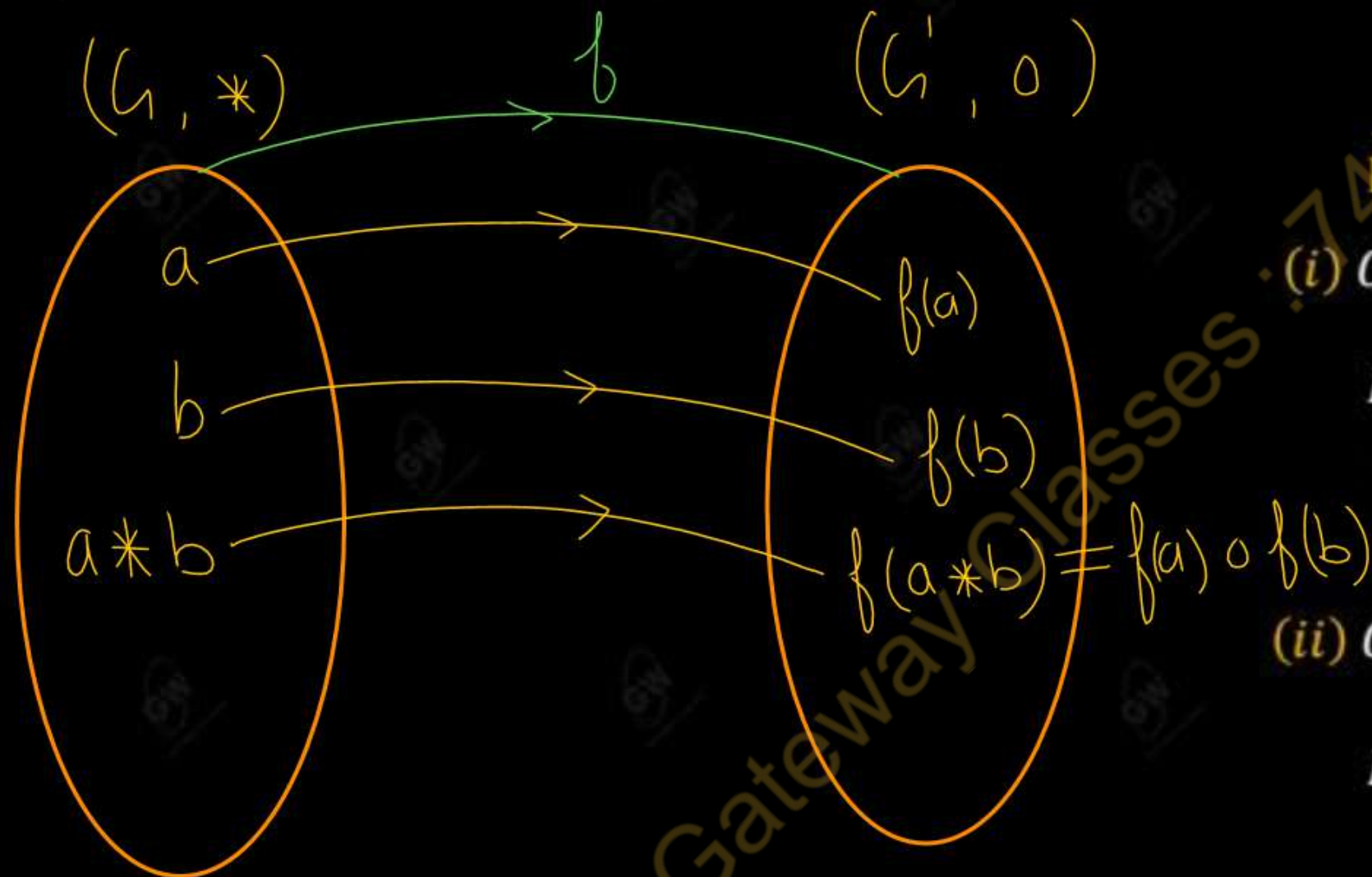
- *Group Homomorphism*
- *Ring*
- *Integral Domain and Field*
- DPP
- PYQ

Let $(G, *)$ and (G', o) be any two groups.

Then a mapping $f: G \rightarrow G'$ is called a Group Homomorphism from G to G' if

$$f(a * b) = f(a) o f(b)$$

$$\forall a, b \in G \text{ and } f(a) o f(b) \in G'$$



Examples

(i) Consider two groups $(G, +)$ and $(G', +)$

If a mapping $f: G \rightarrow G'$ is Homomorphism. Then

$$f(a + b) = f(a) + f(b)$$

$$\forall a, b \in G$$

(ii) Consider two groups $(G, +)$ and $(G', \cdot)$

If a mapping $f: G \rightarrow G'$ is Homomorphism. Then

$$f(a + b) = f(a) \cdot f(b)$$

$$\forall a, b \in G$$

Q.1 Let G be a group of integers under addition and $G = G'$ and let $f(x) = 5x \forall x \in G$ then prove that f is homomorphism.

Given Groups

$(G, +)$ and $(G', +)$

Let $a, b \in G$

$\Rightarrow a+b \in G$

$$f(a) = 5a$$

$$f(b) = 5b$$

$$f(a+b) = 5(a+b)$$

$$f(a+b) = 5a + 5b$$

$$f(a+b) = f(a) + f(b)$$

$\therefore f$ is Homomorphism

Q.2 Prove that if G be a group of all real numbers under addition and let G' be the group of non zero real numbers under multiplication and $f : G \rightarrow G'$ defined by $f(a) = 2^a \forall a \in G$ then it is a

Group Homomorphism.

Given Groups

$(G, +)$ and $(G', \cdot)$

Let $a, b \in G$

$\Rightarrow a+b \in G$

$$f(a) = 2^a$$

$$f(b) = 2^b$$

$$f(a+b) = 2^{a+b}$$

$$f(a+b) = 2^a \cdot 2^b$$

$$f(a+b) = f(a) \cdot f(b)$$

$\therefore f$ is Homomorphism

Q.3 If for a group G ,
 G is abelian.

Given groups
 $(G, \cdot)$ and $(G', \cdot)$

Let $a, b \in G$

$\Rightarrow a \cdot b \in G$

$$f(a) = a^2$$

$$f(b) = b^2$$

$$f(ab) = (ab)^2$$

$f : G \rightarrow G'$ is given by $f(x) = x^2 \forall x \in G$ is a homomorphism, prove that

$\therefore f$ is Homomorphism

$$f(a \cdot b) = f(a) \cdot f(b)$$

$$(ab)^2 = a^2 b^2$$

$$(ab)(ab) = a a b b$$

$$a(ba)b = a(ab)b \quad \{ \text{By Associative law} \}$$

$$ba = ab \quad \{ \text{By left and Right cancellation law} \}$$

$\therefore$ Satisfied commutative property

Hence, G is abelian

Let $f : G \rightarrow G'$ be a Homomorphic mapping

(i) The f – image of the identity element e of G is the identity element e' of G' i.e

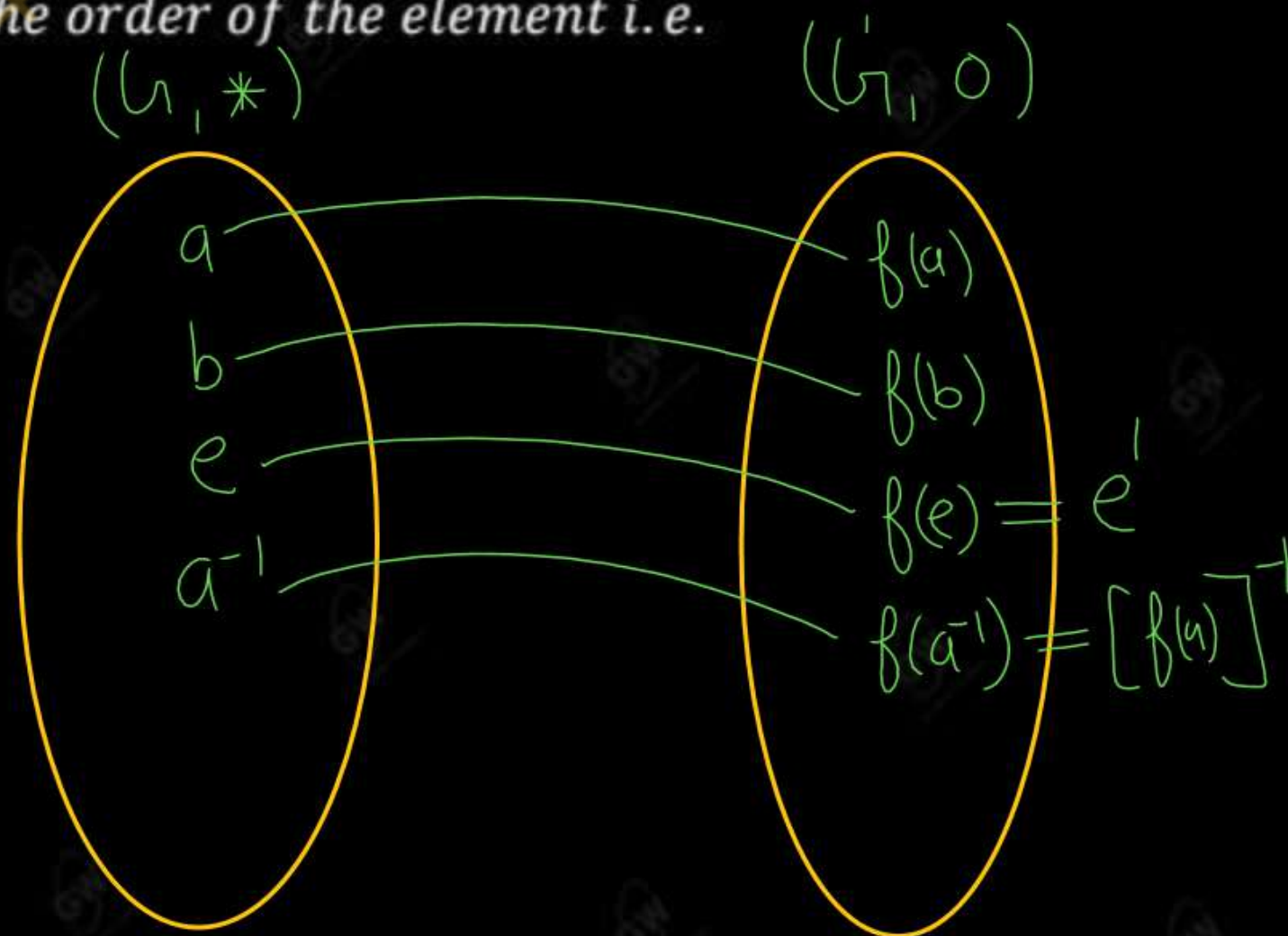
$$f(e) = e'$$

(ii) The f – image of the inverse of an element a of G is the inverse of the f – image of a i.e.

$$f(a^{-1}) = [f(a)]^{-1}$$

(iii) The order of the f – image of an element is the same as the order of the element i.e.

$$o(a) = o[f(a)]$$



Theorem

Let $(G, *)$ and $(G', *')$ be any two groups and let e and e' be their respective identities.

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If f is a homomorphism of G into G' , then prove that

(i) $f(e) = e'$

(ii) $f(x^{-1}) = [f(x)]^{-1} \quad \forall x \in G$

(iii) $o(x) = o[f(x)] \quad \forall x \in G$

PROOF

(i) Let e and e' be the identity element of G and G'

Let $x \in G \Rightarrow f(x) \in G'$

Then

$$e' \cdot f(x) = f(x)$$

$$e' \cdot f(x) = f(ex) \quad \left\{ \because ea = a \right\}$$

$$e' \cdot f(x) = f(e) f(x) \quad \left\{ \because f \text{ is Homomorphic} \right\}$$

$$e' \cdot \cancel{f(x)} = f(e) \cancel{f(x)} \quad \left\{ \text{By Right cancellation law} \right\}$$

$$e' = f(e)$$

$$\Rightarrow \boxed{f(e) = e'}$$

Hence proved

(ii) Let e be the identity element of G and $f(e) = e'$ be the identity element of G' .
Let x^{-1} is the inverse of x in G

$$\Rightarrow x^{-1}x = e = xx^{-1} \quad \text{--- ①}$$

Now

$$x^{-1}x = e$$

$$\Rightarrow f(x^{-1}x) = f(e)$$

$$\Rightarrow f(x^{-1}x) = e'$$

$$\Rightarrow f(x^{-1})f(x) = e' \left\{ \because f \text{ is Homomorphic} \right\}$$

$$f(x^{-1})f(x) = e$$

$$\Rightarrow f(x^{-1}) = [f(x)]^{-1} \left\{ \begin{array}{l} \because ab = e \\ \Rightarrow b = a^{-1} \\ \Rightarrow a = b^{-1} \end{array} \right.$$

$$f(x^{-1}) = [f(x)]^{-1}$$

Hence proved

(iii) Let e be the identity element of G and $f(e) = e'$ be the identity element of G'

Let $x \in G \Rightarrow f(x) \in G'$

Also

Let $o(x) = n$ and $o[f(x)] = m$

Now

$$o(x) = n$$

$$\Rightarrow x^n = e$$

$$\Rightarrow x \cdot x \cdot x \dots n \text{ times} = e$$

$$\Rightarrow f(x \cdot x \cdot x \dots n \text{ times}) = f(e)$$

$$\Rightarrow f(x) f(x) f(x) \dots n \text{ times} = e'$$

$$\Rightarrow [f(x)]^n = e'$$

$$\Rightarrow o[f(x)] \leq n$$

$$\Rightarrow m \leq n \quad \text{--- (1)}$$

Also

$$o[f(x)] = m$$

$$\Rightarrow [f(x)]^m = e'$$

$$\Rightarrow f(x) f(x) f(x) \dots m \text{ times} = f(e)$$

$$\Rightarrow f(n, n, n, \dots \text{---} m \text{ times}) = f(e)$$

$$\Rightarrow n, n, n, \dots \text{---} m \text{ times} = e$$

$$\Rightarrow n^m = e$$

$$\Rightarrow O(n) \leq m$$

$$\Rightarrow n \leq m \text{ --- } (2)$$

From ① and ②

$$m = n$$

$$\Rightarrow O(n) = O[f(n)]$$

Hence proved

(i) Monomorphism

A Homomorphism $f : G \rightarrow G'$ is called monomorphism, if f is **one – one** (injection)

(ii) Epimorphism

A Homomorphism $f : G \rightarrow G'$ is called Epimorphism, if f is **onto** (Surjection)

(iii) Isomorphism

A Homomorphism $f : G \rightarrow G'$ is called Isomorphism, if f is

(i) **one – one**

(iii) **onto**

i.e. bijection

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(iv) *Endomorphism*

A Homomorphism $f : G \rightarrow G'$ is called endomorphism, if $G' = G$

(v) *Automorphism*

A Homomorphism $f : G \rightarrow G'$ is called Automorphism, if $G' = G$ and f is one – one
onto (bijection)

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An algebraic structure $(R, +, \cdot)$ is called ring if

(i) $(R, +)$ is an Abelian group

(ii) $(R, \cdot)$ is a Semi group

(iii) R satisfied Distributive law

$$(1) a \cdot (b + c) = a \cdot b + a \cdot c$$

{Left distributive law}

$$(2) (b + c) \cdot a = b \cdot a + c \cdot a$$

{Right distributive law}

Ring with Unity

A ring $(R, +, \cdot)$ is called ring with unity if $\forall a \in R \exists$ an identity element $1 \in R$ such that

$$a \cdot 1 = a = 1 \cdot a$$

The element $1 \in R$ is called unit element of the ring

Ring without Unity

A ring $(R, +, \cdot)$ which does not contain multiplicative identity is called a ring without unity

Commutative Ring or Abelian Ring

A ring $(R, +, \cdot)$ is said to be a commutative ring or an abelian ring if it satisfied commutative law

$$\text{i.e. } \forall a, b \in R$$

$$a \cdot b = b \cdot a$$

Commutative Ring with Unity

If a ring $(R, +, \cdot)$ is called commutative ring with unity if it satisfied

- (i) Commutative law under multiplication
- (ii) Identity law under multiplication

If R is a ring, then $\forall a, b, c \in R$

PROOF

(i) $a \cdot 0 = 0 \cdot a = 0$

(ii) $a(-b) = -(ab) = (-a)b$

(iii) $(-a)(-b) = ab$

(iv) $a(b - c) = ab - bc$

(v) $(b - c)a = ba - ca$

(i) $a0 = 0a = 0$

$$a0 = a(0+0) \left\{ \because 0+0=0 \right\}$$

$$0+a0 = a(0+0) \left\{ \because 0+a=a \right\}$$

$$0+a0 = a0+a0 \left\{ \text{By Left Distributive law} \right\}$$

$$0+\cancel{a0} = a0+\cancel{a0} \left\{ \text{By Right cancellation law} \right\}$$

$$0 = a0$$

$$\Rightarrow \boxed{a0 = 0} \text{ Hence proved}$$

Also

$$0a = (0+0)a \quad \{ \because 0+0=0 \}$$

$$0+0a = (0+0)a \quad \{ \because 0+a=a \}$$

$$0+0a = 0a+0a \quad \{ \text{By Right Distributive Law} \}$$

$$0+\cancel{0a} = 0a+\cancel{0a} \quad \{ \text{By Right cancellation Law} \}$$

$$0 = 0a$$

$$\Rightarrow \boxed{0a = 0}$$

$$(ii) a(-b) = -(ab) = (-a)b$$

We know that

$$a0 = 0$$

$$a(-b+b) = 0 \quad \{ \because -b+b=0 \}$$

$$a(-b) + ab = 0 \quad \{ \text{By Left Distributive law} \}$$

$$\boxed{a(-b) = -(ab)}$$

Similarly

$$0b = 0$$

$$(-a+a)b = 0 \quad \{ \because -a+a=0 \}$$

$$(-a)b + ab = 0 \quad \{ \text{By Right Distributive law} \} \Rightarrow \boxed{(-a)(-b) = ab}$$

$$\boxed{(-a)b = -(ab)}$$

$$\therefore \boxed{a(-b) = -(ab) = (-a)b}$$

$$(III) \quad (-a)(-b) = ab$$

LHS

$$(-a)(-b) = -a(-b)$$

$$= -[a(-b)]$$

$$= -[-(ab)] \quad \{ a(-b) = -(ab) \}$$

$$= ab$$

$$(IV) \quad a(b-c) = ab - bc$$

LHS

$$a(b-c) = a[b + (-c)]$$

$$a(b-c) = ab + a(-c) \quad \{ \text{By Distributive law} \}$$

$$\boxed{a(b-c) = ab - ac} \quad \{ \because a(-c) = -ac \}$$

$$(V) \quad (b-c)a = ba - ca$$

LHS

$$(b-c)a = [b + (-c)]a$$

$$(b-c)a = ba + (-c)a \quad \{ \text{By Distributive law} \}$$

$$\boxed{(b-c)a = ba - ca}$$

Ring with zero divisor

A ring $(R, +, \cdot)$ is called a ring with zero divisor if $\forall a, b \in R$

$$ab = 0 \Rightarrow a \neq 0, b \neq 0$$

Ring without zero divisor

A ring $(R, +, \cdot)$ is called a ring without zero divisor if $\forall a, b \in R$

$$ab = 0 \Rightarrow a = 0 \text{ or } b = 0$$

$$ab \neq 0 \Rightarrow a \neq 0 \text{ and } b \neq 0$$

Integral Domain

A ring $(R, +, \cdot)$ is called an integral domain if

- (i) R is commutative
- (ii) R has an identity element
- (iii) R is without zero divisors

$$3 \times_6 2 = 0$$

Field

A ring $(F, +, \cdot)$ with at least two elements is called field if

- (i) F is Commutative
- (ii) F has identity element
- (iii) Each non zero element of F have multiplicative inverse

OR

A ring $(F, +, \cdot)$ with at least two elements is called field if

- (i) $(F, +)$ is an Abelian group
- (ii) $(F, \cdot)$ is an Abelian group

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Topic : Ring, Integral Domain and Field

Q. 1 Prove that the set of integers Z with respect to $+$ and $\cdot$ is a ring.

Q. 2 Prove that $(I, +, \cdot)$ is an integral domain

Q. 3 Prove that $(R, +, \cdot)$ is a Field $(R, +)$ $(R, \cdot)$

Q. 4 Prove that $(Q, +, \cdot)$ is a Field

Q. 5 Prove that $(I, +, \cdot)$ is not a Field

Q. 6 Prove that the set of residues $F = \{0, 1, 2, 3, 4\}$ modulo 5 is a field wrt addition

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and multiplication of residue classes modulo 5 i.e. $(f, +_5, \times_5)$ is a Field

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Thank You